SuperRPCA: A Collaborative Superpixel Representation Prior-Aided RPCA for Hyperspectral Anomaly Detection

Jhao-Ting Lin, *Student Member, IEEE*, and Chia-Hsiang Lin, *Member, IEEE*

***Abstract*— Recently, numerous hyperspectral anomaly detection (HAD) methods have been proposed for broad and crucial applications. Among these, robust principal component analysis (RPCA) gains considerable attention in HAD, as it separates the matrix into global low-rank (LR) and sparse components, corresponding to the property of background and anomaly. However, RPCA solves the problem by treating hyperspectral imagery (HSI) as a matrix, but this approach alone cannot well-describe the local spatial texture information. In addition, a pixelwise detection method, collaborative representation detector (CRD), has been proposed, which exploits the vital piece of local information by assuming that background pixels can be composed of their neighbor pixels, while abnormal ones cannot. Although several CRD-based methods achieve promising HAD performances, they generally suffer from high computational costs due to the pixelwise optimization scheme. To overcome the aforementioned two limitations, we propose a novel algorithm, SuperRPCA. First, we improve CRD to superpixelwise calculation and reconstruct the background with a simplex-based algebraic solution. Subsequently, the rebuilt background is tailored to serve as a convex regularizer and integrated into RPCA. Besides, the regularizer inherently possesses an LR property, adeptly substituting the nuclear norm in traditional RPCA and hence significantly reducing computational costs. SuperRPCA demonstrates easily identifiable visual qualities and state-of-the-art quantitative performance with all the closed-form algorithmic expressions explicitly derived.**

***Index Terms*— Collaborative representation detector (CRD), convex optimization, hyperspectral anomaly detection (HAD), robust principal component analysis (RPCA), superpixel segmentation.**

I. Introduction

With the rapid advancement of technology, the space remote sensing industry has flourished over the past few decades. Within remote sensing, satellite images can be categorized into multispectral images (MSIs) and hyperspectral images (HSIs) according to coarse and fine spectral resolution over contiguous ranges of wavelengths, respectively. The superior spectral resolution of HSIs results in a more continuous spectral reflectance signature. In particular, the abundance of spectral information in HSIs offers numerous critical applications, such as land use and land cover (LULC) change detection, environmental monitoring, material identification, and target/anomaly detection [1], [2], which are challenging to achieve with traditional RGB images.

Among the applications above, hyperspectral target/anomaly detection (HTD/HAD) has garnered considerable attention in various civilian and military applications [3], [4] due to its ability to identify specific materials or objects in HSIs. HTD is fundamentally a binary classification task, where the goal is to distinguish the target of interest from the unwanted background. The nature of the target can exhibit significant diversity, encompassing from rare minerals to aircraft, for instance. In addition, the spectral signature of the target can be obtained beforehand or stored in a database. This enables valuable prior knowledge to be leveraged in identifying the target. However, due to satellite hardware limitations and financial considerations, HSI typically suffers from coarse spatial resolution. Consequently, various signatures may mix within a single pixel [2], hindering accurate target identification in HTD. In addition, traditional HTD methods rely on the target's spectral signature, making them unsuitable for scenarios where this information is unavailable. This limitation is particularly critical in real-world applications such as battlefield target detection. To tackle this challenge, HAD techniques are explored to identify the pixels based on their deviation from the background. Apart from the remote sensing field, anomaly detection has also attracted much attention in various areas. For instance, to enhance passenger satisfaction and public safety, [5] analyzed the movement patterns of passengers to apprehend pickpockets in public transportation systems. To forestall wireless sensor networks (WSNs) from causing harm to networks, [6] proposed certain mechanisms to serve as a more comprehensive protection wall.

In recent decades, statistical-based methods have been extensively studied by scholars. The Reed–Xiaoli algorithm (RX) [7] is considered one of the benchmark HAD algorithms. RX assumes that the statistical properties of the background follow a Gaussian distribution, where the mean and covariance are estimated from the available data. The algorithm then calculates the Mahalanobis distance [8] between the pixel under test (PUT) and the data centroid in multivariate space.

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Jhao-Ting Lin is with the Institute of Computer and Communication Engineering, Department of Electrical Engineering, National Cheng Kung University, Tainan 701, Taiwan (e-mail: q38091534@gs.ncku.edu.tw).

Chia-Hsiang Lin is with the Department of Electrical Engineering and the Miin Wu School of Computing, National Cheng Kung University, Tainan 701, Taiwan (e-mail: chiahsiang.steven.lin@gmail.com).

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The closer/further the distance, the more likely the PUT is a background/anomaly pixel. Since natural scenes rarely conform to this assumption, various extension methods have been proposed based on RX. In contrast to RX, local RX (LRX) [9] solely gathers pixels surrounding the PUT for adapting the rapidly varying scenes in the data. Considering that the anomaly pixels are grouped with a specific size, LRX introduces a double concentric sliding window to prevent collection contamination. Kwon and Nasrabadi [10] extended RX by combining the kernel method, which is commonly used in machine learning. The kernel method projects the data from the original input space to a higher dimensional feature space through a nonlinear mapping, making it easier to discriminate between background and anomaly, particularly when the data are complicated. Other enhanced versions of RX, such as minimum covariance determinant RX, subspace RX, and causal RX, can be referred to [11], [12], [13]. However, a major challenge of statistical-based methods lies in the unknown nature of the probability density functions in real-world scenarios. Hence, [14] used signal-to-noise ratio (SNR) as an alternative criterion for designing a detector. Furthermore, the method is the first dual theory of HTD developed for HAD.

Apart from the statistical-based approaches, some scholars have explored the matrix decomposition algorithms. Based on low-rank and sparse matrix decomposition (LRaSMD) [15], an HSI can be expressed as a combination of a low-rank (LR) matrix and a sparse matrix that conforms to the characteristics of the background and anomalies, respectively. Chen et al. [16] formulated an optimization problem based on robust principal component analysis (RPCA) [17]. The primary objective of this problem lies in the restoration of an LR matrix from data severely compromised by sparse noise. A more detailed discussion of RPCA will be provided in Section II. Considering the issue of noise in real-world HSIs, Zhang et al. [18] adopted a rapid approximation algorithm called Go decomposition (GoDec) [19] to solve the HAD problem meanwhile removing the mixed noise. In the GoDec optimization, two matrices are alternatively approximated to LR and sparse attributes, respectively, whose concept is inspired by the behaviors of two players in the game of Go. Unlike previously mentioned methods of recovering the LR matrix from a single subspace, the LR representation (LRR) [20], [21] extends this concept to multiple subspaces. Specifically, LRR decomposes the LR matrix into a dictionary multiplied by a coefficient matrix and then finds the lowest-rank representation of the coefficient matrix. Building upon the background dictionary construction, Huyan et al. [22] further constructed the other dictionary tailored to capture the characteristics of potential anomalies. However, this approach is time-consuming by two dictionary constructions. Based on LRR, Xu et al. [23] proposed LR and sparse representation (LRASR), which assumes that the coefficient matrix not only has an LR property but also sparsity to better characterize local spatial structure. In addition, graph and total variation (TV) regularized LRR (GTVLRR) [24] and abundance- and dictionary-based LR decomposition (ADLR) [25] are also proposed based on LRR; the former adopts graph and TV regularization to capture spatial structure while the latter incorporates the LRR into spectral unmixing. When it comes to HAD solved by spectral unmixing, empirical evidence indicates that the identical materials can be both an anomaly and a background at the same time. For instance, a fighter jet and an architectural roof may be constructed from aluminum; however, the detection task pertains solely to identifying the former as the anomaly. Therefore, spectral unmixing-based methods may have an unsatisfactory detection result.

The rapid evolution of artificial intelligence (AI) is expected to impact various fields of science and engineering [26]. Specifically, deep learning's capability to manage nonlinear real-world effects has driven numerous researchers to apply it to HAD, as evidenced by the growing works in this field [27], [28]. For instance, autoencoder (AE) is particularly suitable for HAD tasks due to its effective data representation. Typically, AE is composed of two main components, namely, an encoder and a decoder; the encoder takes in the input data and then maps it to a lower dimensional representation; the decoder, on the other hand, returns the lower dimensional representation to the original data space. One of the notable methods is the autonomous HAD network (Auto-AD) [29], which uses its AE network to reconstruct the background while detecting the anomalies through the identification of reconstruction errors. To further minimize the occurrence of anomalous reconstructions, an adaptive-weighted loss function is designed to reduce the weights of potential anomalous pixels during training. By incorporating AE into the optimization iterations, the deep LR prior-based method (DeepLR) [30] can reduce the responses of anomalies for better distilling the pure background from AE. Although reconstructing anomalies in AE is more challenging than background, the decoder will reinforce anomaly reconstruction if more information is preserved in the encoded features. Therefore, instead of relying solely on features, Huyan et al. [31] replaced them with a cluster-based memory module, which helps avoid the influence of anomalies, enabling the decoder to focus on reconstructing a purer background. Unlike the aforementioned approaches that process the data within the Euclidean space, Sun et al. [32] introduced a hyperbolic space-based AE (HSAE). HSAE leverages the inherent spatial hierarchical structure of data in hyperbolic geometry, effectively reducing anomaly reconstruction. These AE-based methods have shown promising results in HAD, especially for complicated scenes with nonlinear spectral characteristics.

We should also mention that there are other types of HAD approaches, for instance, Gao et al. [33] designed a chessboard-shaped topological framework to disassemble the HSI and mine deep-level features and Li et al. [34] separated the anomalies and background by introducing the isolation forest in the kernel space. These methods leverage unique perspectives, leading to a broader range of solutions for HAD problem. The subsequent sections of this article are organized as follows. In Section II, two HAD methods, RPCA and collaborative representation detector (CRD), will be detailed that are highly related to our work. In Section III, we will integrate a superpixel segmentation method into CRD to reconstruct the background roughly. In Section IV, we discuss

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the motivation behind designing the reconstructed background as a convex regularizer and how it is incorporated into the RPCA model. In addition to demonstrating the superiority of general quantitative and qualitative comparisons, we also provide an analysis of the key parameters in Section V. Finally, we conclude this article in Section VI. Mathematical details are then given in the Appendix.

Before we delve into two HAD methods related to our work, some standard notations are summarized hereinafter for readers to consult conveniently. $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$, denote, respectively, the set of real-valued n-dimensional vectors and the set of real-valued m-by-n matrices. The symbols $\mathbf{1}_m$, $\mathbf{0}_{m\times n}$, and $\boldsymbol{I}_m$ represent, respectively, an all-one column vector of size m, an all-zero matrix of size m-by-n, and an identity matrix of size m-by-m. $\mathcal{I}_N \triangleq \{1,\ldots,N\}$ for any positive integer N. $\boldsymbol{U}[i]$ denotes the ith column of the matrix $\boldsymbol{U}$, and $\boldsymbol{u}[i]$ denotes the ith entry of the vector $\boldsymbol{u}$. For a matrix $\boldsymbol{M}\in\mathbb{R}^{m\times n}$, $\text{rank}(\boldsymbol{M})$ and $\|\boldsymbol{M}\|_*$ denote the rank and the nuclear norm of $\boldsymbol{M}$, respectively, in which $\|\boldsymbol{M}\|_* = \sum_{i=1}^{\min\{m,n\}}\sigma_i(\boldsymbol{M})$, and $\sigma_i(\boldsymbol{M})$ is the ith largest singular value of $\boldsymbol{M}$. The ℓ_p-norm and Frobenius norm are denoted by $\|\cdot\|_p$ and $\|\cdot\|_F$, respectively. $\text{vec}(\cdot)$ denotes the vectorization operator. $\otimes$ denotes the Kronecker product operator. $\succeq$ is the componentwise inequality.

II. Related HAD Works

In this section, we initially present some background information on RPCA (see Section II-A). In addition, we review a classical representation-based technique known as CRD (see Section II-B). Following that, we provide an overview of the proposed method (see Section II-C) that effectively leverages the advantages of both RPCA and CRD while compensating for each other's limitations.

A. Robust Principal Component Analysis

Regarding principal component analysis (PCA) [35], it is widely recognized as a common statistical tool for data analysis. In the remote sensing field, PCA plays a vital role in the highly correlated spectral bands of HSIs. Nevertheless, the performance of PCA may be seriously compromised by arbitrary gross errors, such as outliers that considerably deviate from the data distribution, leading to inaccurate principal components. To tackle this challenge, the RPCA [17], [36] has been proposed to efficiently solve such a conundrum in polynomial time.

Here, we demonstrate how RPCA can be applied to HAD. Initially, a 3-D hyperspectral data cube is first reshaped to 2-D form $\boldsymbol{Y}\in\mathbb{R}^{M\times L}$, where M and L denote the number of bands and pixels, respectively. RPCA assumes that $\boldsymbol{Y} = \boldsymbol{B} + \boldsymbol{A}$, where $\boldsymbol{B}$ denotes an LR matrix and $\boldsymbol{A}$ is a sparse matrix. As discussed in Section I, HAD aims to classify an HSI into background and anomaly parts, whose properties coincidentally correspond to LR and sparse, respectively. Moreover, the anomaly pixels only exist with a low fraction in the data, so it is suitable to model HAD problem as RPCA. The key difference between PCA and RPCA is that while PCA can only tolerate small perturbations, the sparse matrix $\boldsymbol{A}$ in RPCA can accommodate arbitrarily large magnitudes. As a result, RPCA is indeed more robust than PCA in processing highly corrupted data, and such a decomposition can be intuitively solved through the optimization problem as below

$$\begin{aligned}&\min_{\boldsymbol{B},\boldsymbol{A}}\ \text{rank}(\boldsymbol{B}) + \lambda_{\text{RPCA}}\|\boldsymbol{A}\|_0\\ &\text{s.t. } \boldsymbol{Y} = \boldsymbol{B} + \boldsymbol{A}\end{aligned} \tag{1}$$

where $\lambda_{\text{RPCA}} > 0$ is the regulairzer parameter. However, both the terms in (1) are nonconvex, leading to the problem being NP-hard to solve. To overcome this challenge, an alternative is to use convex relaxation techniques to find a convex surrogate, as demonstrated below

$$\begin{aligned}&\min_{\boldsymbol{B},\boldsymbol{A}}\ \|\boldsymbol{B}\|_* + \lambda_{\text{RPCA}}\|\boldsymbol{A}\|_1\\ &\text{s.t. } \boldsymbol{Y} = \boldsymbol{B} + \boldsymbol{A}.\end{aligned} \tag{2}$$

The nuclear norm and the ℓ_1-norm represent the convex envelopes of rank and ℓ_0-norm, respectively. Although (2) can be directly solved by the interior point method (IPM) [37], IPM can only handle small-scale optimization problems due to the cumbersome Hessian matrix calculation. Therefore, the first-order optimization solvers, such as accelerated proximal gradient (APG) methods [38] and the augmented Lagrange multiplier (ALM) [39] algorithm, are proposed to avoid laborious calculation. During primal variable updating stage in ALM, nuclear norm and ℓ_1-norm can be solved by singular value thresholding (SVT) [40] and the shrinkage method, respectively, and both of them have simple closed-form solutions.

B. Collaborative Representation Detector

Unlike the matrix decomposition approaches, CRD [41] is a representation-based method that performs the calculation pixelwisely. For each PUT, the detector customizes different local dictionaries consisting of surrounding pixels to represent it. Due to the background's LR nature, CRD assumes that the background pixels can be easily represented, while anomaly pixels cannot. Given an HSI $\boldsymbol{Y}\in\mathbb{R}^{M\times L}$, each PUT is denoted by $\boldsymbol{y}_i\in\mathbb{R}^M, \forall i\in 1,\ldots,L$. The general objective function can be formulated as a data-fitting term with ridge regression (also known as Tikhonov regularization) as follows:

$$\arg\min_{\boldsymbol{\alpha}_i}\ \|\boldsymbol{y}_i - \boldsymbol{N}_i\boldsymbol{\alpha}_i\|_2^2 + \lambda_{\text{CRD}}\|\boldsymbol{\alpha}_i\|_2^2 \tag{3}$$

where $\boldsymbol{N}_i\in\mathbb{R}^{M\times P}$ is the set that collects P spatial neighbor pixels of $\boldsymbol{y}_i$, $\boldsymbol{\alpha}_i\in\mathbb{R}^P$ is the corresponding weight, and $\lambda_{\text{CRD}} > 0$ is a balanced parameter. Just like LRX, CRD also adopts the dual window strategy to avoid selecting anomaly pixels. After obtaining the optimized weight $\boldsymbol{\alpha}_i^\star$, a residual $\boldsymbol{r}\in\mathbb{R}^L$ can be calculated between $\boldsymbol{y}_i$ and $\hat{\boldsymbol{y}}_i$ (i.e., reconstructed $\boldsymbol{y}_i$), which can be expressed as follows:

$$\boldsymbol{r}[i] = \|\boldsymbol{y}_i - \hat{\boldsymbol{y}}_i\|_2 = \left\|\boldsymbol{y}_i - \boldsymbol{N}_i\boldsymbol{\alpha}_i^\star\right\|_2. \tag{4}$$

Intuitively, when the residual value of $\boldsymbol{r}[i]$ is either small or large, it indicates that $\boldsymbol{y}_i$ is more likely to be a background or anomaly, respectively.

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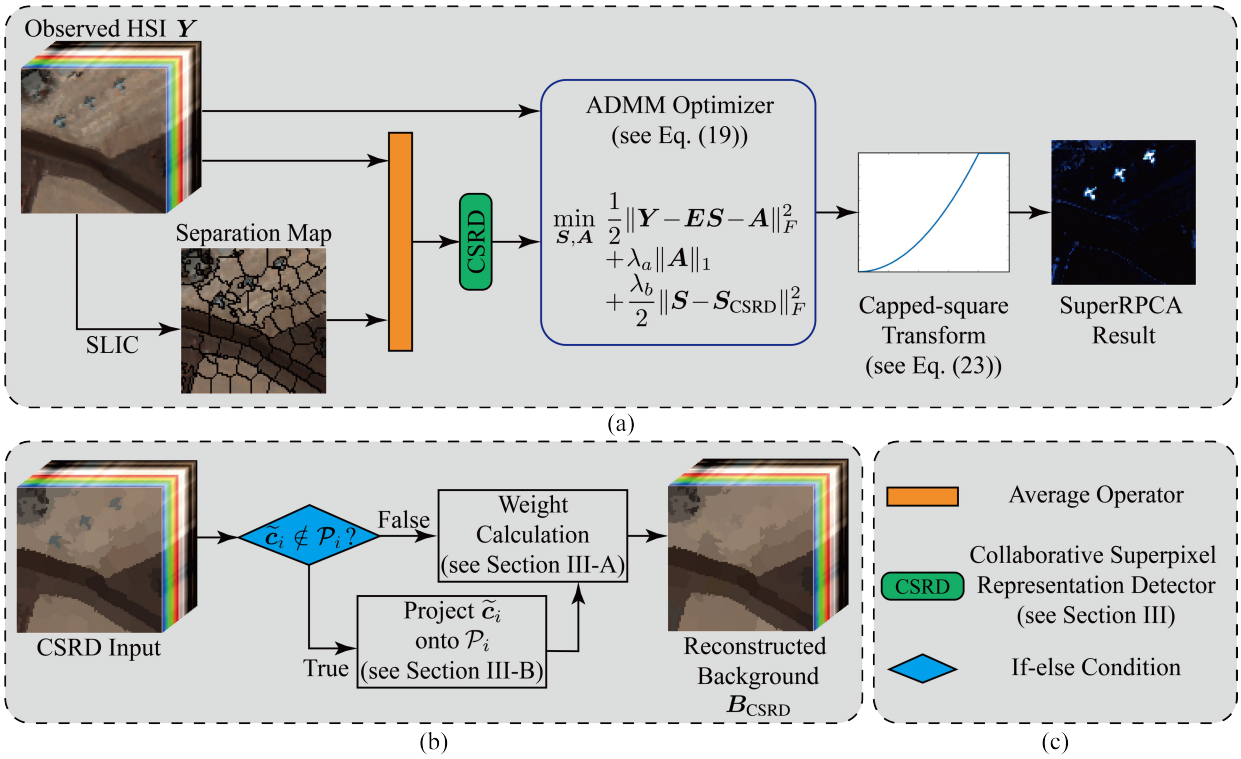


Fig. 1. Overview of the proposed (a) SuperRPCA and (b) CSRD. (c) Related annotations.

C. Overview and Contribution

While RPCA effectively decomposes HSIs into LR and sparse components, this matrix-based method struggles with using valuable local spatial texture information. On the other hand, although CRD-based methods perform better than traditional HAD methods through their local representation assumptions, they require computationally expensive pixel-level computations, thus hindering their practicality. To address these issues, we introduce a novel algorithm that leverages the benefits of both the approaches. First, we develop a collaborative superpixel representation detector (CSRD) by incorporating CRD with a superpixel segmentation algorithm called simple linear iterative clustering (SLIC) [42], allowing it to exploit local spatial texture information while maintaining computational efficiency. However, previous works [41], [43] only promote the sum-to-one property on the representation weights, which can be inadequate to deal with more comprehensive data. To allow weights satisfying sum-to-one and nonnegative constraints, which strictly conform to the fully constrained least squares (FCLS) problem [44], we propose a simplex-based algebraic solution. While CSRD enables rapid background reconstruction due to its efficient local representation, the utilization of SLIC will introduce a mosaic-like appearance in the reconstructed background. To fully harness this information, the reconstructed background is skillfully customized as a convex regularizer and integrated into the RPCA optimization problem. Remarkably, the regularizer effectively substitutes the nuclear norm in the RPCA model, leading to a significant reduction in computational costs associated with the SVT step. The overall algorithm is graphically summarized in Fig. 1(a), CSRD is illustrated in Fig. 1(b), and the related legends are presented in Fig. 1(c). The novelties and the critical contributions of this work can be concisely summarized as follows

1) Unlike the pixelwise calculation mechanism of the traditional CRD, we introduce the CSRD mechanism that dramatically alleviates the computational burden, as it deals with a tractable set of K superpixels instead of L individual pixels ($K \ll L$). Besides the above trick, the proposed algorithm, called SuperRPCA, has the convergence guarantee and has all its algorithmic steps derived with closed-form expressions using elegant convex geometry properties.
2) The proposed SuperRPCA leverages the local neighbor information from CSRD's superpixelwise calculation and the global information from the RPCA framework. This hierarchical strategy is proven to be effective to capture the global–local geometric structure of the target HSI, thereby achieving effective and robust detection results.
3) Our customized CSRD-based regularizer avoids the commonly used nuclear-norm regularizer in the background removal literature, thus avoiding the time-consuming SVT computing stage, while still keeping an efficient background recovery capability. All in all, even without relying on the big data training stage of typical data-driven deep learning detection methods, the proposed SuperRPCA achieves state-of-the-art HAD performance (see Section V).

III. Collaborative Superpixel Representation Detector

Relying on empirical observations, anomaly objects do not occupy the scene in a single pixel; instead, they often

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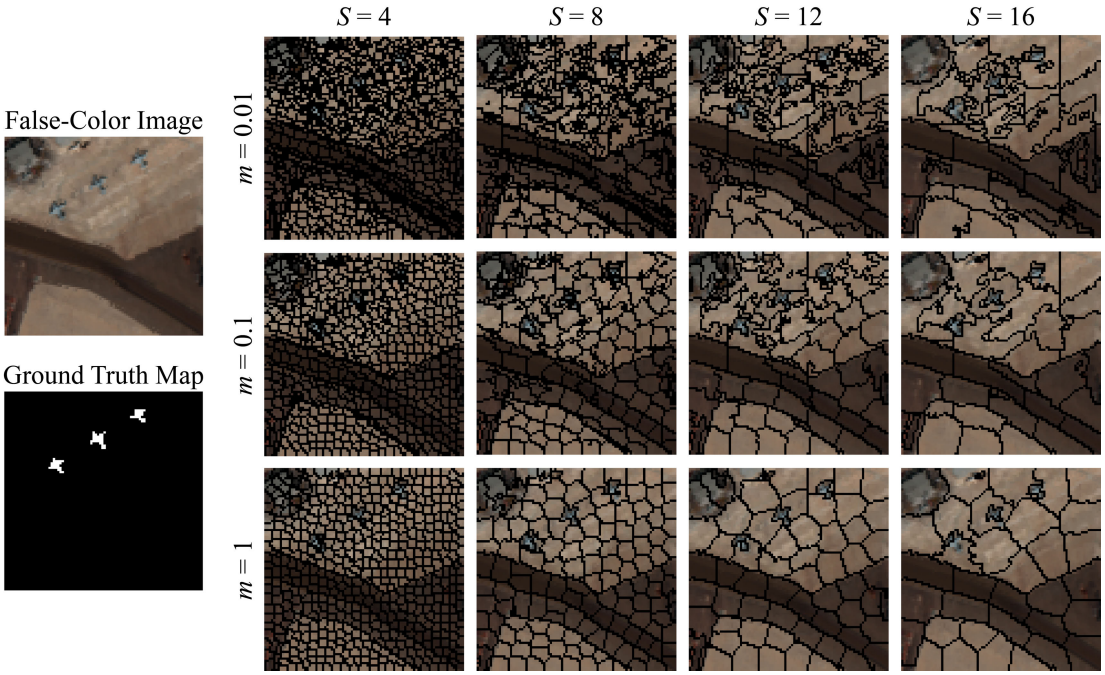


Fig. 2. SLIC allows for the configuration of two crucial parameters, namely, S and m. The parameter S controls the size of the superpixels and is directly responsible for determining their number. On the other hand, the parameter m regulates the superpixel appearance.

cluster adjacent pixels into a group with irregular appearance. Motivated by this fact, we adopt a superpixel segmentation technique, SLIC[1] [42], as a preprocessing step. SLIC is exploited based on the K-means clustering [45]; in addition to measuring the similarity in spectral characteristics among pixels, SLIC also takes into account the spatial context. Two vital parameters, S and m, can be configured in SLIC, where S determines the superpixel size and directly affects the number of superpixels, while m controls the superpixel appearance. Specifically, we can achieve a more grid-like segmentation or allow each superpixel to follow the local boundary more closely by adjusting m. The influence between the two parameters is illustrated in Fig. 2. Due to the high redundancy between the HSI bands, for a given HSI $\boldsymbol{Y} \in \mathbb{R}^{M\times L}$, SLIC is applied to the first three principal components after PCA [35]. By appropriately setting two parameters S and m, we obtain a segmentation map, in which $\boldsymbol{Y}$ is clustered into K superpixels ($K \ll L$), and each superpixel is denoted by $\boldsymbol{Y}_i \in \mathbb{R}^{M\times L_i}$, $\forall i \in \mathcal{I}_K$ with $\sum_{i=1}^{K} L_i = L$; pixels among a superpixel have distinct but similar properties, such as close pixel coordinates and homogeneous spectra. Here, we assume that the superpixels can be classified into background and anomaly parts.

After the preprocessing step, we can bridge the connection between the original pixel space and the superpixel one by calculating the mean of the spatial coordinates and spectral information for all the pixels in $\boldsymbol{Y}_i$, denoted by $\boldsymbol{l}_i \in \mathbb{R}^2$ and $\boldsymbol{c}_i \in \mathbb{R}^M$, respectively. For convenience, these two vectors are concatenated to form spatial–spectral superpixel centroids (SCs) $\boldsymbol{V} = [\boldsymbol{L}^{\mathrm{T}}, \boldsymbol{C}^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{(M+2)\times K}$, in which $\boldsymbol{L} = [\boldsymbol{l}_1, \ldots, \boldsymbol{l}_K] \in \mathbb{R}^{2\times K}$ and $\boldsymbol{C} = [\boldsymbol{c}_1, \ldots, \boldsymbol{c}_K] \in \mathbb{R}^{M\times K}$. So far, the ith SC can appropriately represent the pixels in $\boldsymbol{Y}_i$,

[1]This work implements SLIC by an open library of computer vision algorithms: https://www.vlfeat.org/

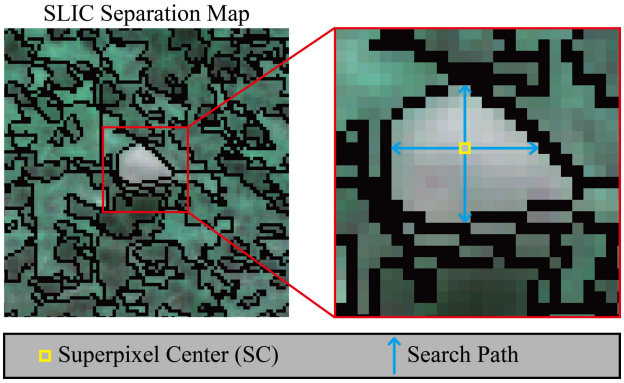


Fig. 3. For the ith SC, the collection mechanism regards $\boldsymbol{l}_i$ as the data coordinates and then searches through along with the four directions until touching with the other superpixel $\boldsymbol{Y}_j$, $j \neq i$. For instance, the search is initiated from the pixel marked with a yellow border and then follows four blue arrows until it reaches endpoints.

allowing us to perform fewer calculations on $\boldsymbol{V}$ rather than the original HSI cube $\boldsymbol{Y}$.

Inheriting the assumption of CRD, if the ith SC belongs to the background/anomaly, its spectrum $\boldsymbol{c}_i$ would be easily/difficultly represented by a linear combination among the set of adjacent SCs' spectra $\boldsymbol{P}_i = \{[\boldsymbol{c}_1, \ldots, \boldsymbol{c}_{N_i}] \mid \boldsymbol{c}_j \in \boldsymbol{C},\ j \neq i\} \in \mathbb{R}^{M\times N_i}$, where N_i denotes the number of the adjacent SCs. For how to collect the adjacent SCs, we design a search mechanism based on the first-order neighborhood.[2] In detail, we adopt $\boldsymbol{l}_i$ as the datum coordinates and search through along with the four directions until osculating the other superpixels $\boldsymbol{Y}_j$, $j \neq i$, as shown in Fig. 3. The repeatedly collected SCs will be automatically removed to

[2]The first-order neighborhood is denoted by the four closest pixels of the PUT (i.e., up, down, left, and right directions) [46].

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ensure that each adjacent SC is independent of the others. Our neighborhood collection strategy is unaffected when different scale anomalies appear in the scenario. This is attributed to the utilization of SLIC, which adaptively characterizes anomaly superpixel appearance, instead of using fixed-size dual windows to collect neighboring pixels like CRD.

In the previous work [43], it only promoted the sum-to-one property of the corresponding weights, which may result in a tremendous positive or negative value of each weight, thereby reducing the detection accuracy in some scenes. Just like several hyperspectral unmixing (HU) problems that regard each pixel as a mixture of multiple hyperspectral signatures through linear mixing model (LMM) [2], [47], we consider each SC to be a convex combination of its neighboring SCs. Therefore, for each SC's spectrum $\boldsymbol{c}_i$, the sum-to-one and nonnegative constraints are imposed into the optimization problem, which is designed as

$$\begin{aligned} &\min_{\boldsymbol{w}_i} \ \|\boldsymbol{c}_i - \boldsymbol{P}_i \boldsymbol{w}_i\|_2^2 \\ &\text{s.t. } \mathbf{1}_{N_i}^{\mathrm{T}} \boldsymbol{w}_i = 1, \quad \boldsymbol{w}_i \succeq \mathbf{0} \end{aligned} \tag{5}$$

where $\boldsymbol{w}_i \in \mathbb{R}^{N_i}$ denotes the weights corresponding to the collected SCs in $\boldsymbol{P}_i$. To deal with the constraints in (5), our intuition is to solve the problem from a geometric perspective. Specifically, if we can find the optimal $\boldsymbol{w}_i$ which belongs to the simplex constructed by $\boldsymbol{P}_i$, both the constraints can be satisfied directly. Primarily, we approximately represent all SCs' spectra $\boldsymbol{C}$ in the $(N_i - 1)$-dimension using PCA, i.e.,

$$\boldsymbol{C} \approx \boldsymbol{F}\widetilde{\boldsymbol{C}} + \boldsymbol{M} \tag{6}$$

where $\boldsymbol{F} \in \mathbb{R}^{M \times (N_i - 1)}$ forms an orthonormal basis of the subspace (i.e., $\boldsymbol{F}^{\mathrm{T}}\boldsymbol{F} = \boldsymbol{I}_{N_i - 1}$), $\boldsymbol{M} \in \mathbb{R}^{M \times K}$ is offset which moves data distribution to zero mean, and $\widetilde{\boldsymbol{C}} = [\widetilde{\boldsymbol{c}}_1, \ldots, \widetilde{\boldsymbol{c}}_K] \in \mathbb{R}^{(N_i - 1) \times K}$ represents $\boldsymbol{C}$ in the dimension reduction (DR) space.

Similarly, in the DR space, the spectrum set of the adjacent SC can be denoted as $\widetilde{\boldsymbol{P}}_i = \{[\widetilde{\boldsymbol{c}}_1, \ldots, \widetilde{\boldsymbol{c}}_{N_i}] \mid \widetilde{\boldsymbol{c}}_j \in \widetilde{\boldsymbol{C}}, \ j \neq i\} \in \mathbb{R}^{(N_i - 1) \times N_i}$, and (5) can be reformulated as

$$\begin{aligned} &\min_{\boldsymbol{w}_i} \ \|\widetilde{\boldsymbol{c}}_i - \widetilde{\boldsymbol{P}}_i \boldsymbol{w}_i\|_2^2 \\ &\text{s.t. } \mathbf{1}_{N_i}^{\mathrm{T}} \boldsymbol{w}_i = 1, \quad \boldsymbol{w}_i \succeq \mathbf{0}. \end{aligned} \tag{7}$$

Since $\widetilde{\boldsymbol{P}}_i$ is capable of forming an $(N_i - 1)$-dimensional simplex $\mathcal{P}_i$ with N_i vertices (i.e., $\mathcal{P}_i = \mathbf{conv}\{\widetilde{\boldsymbol{P}}_i\}$), where $\mathbf{conv}\{\cdot\}$ defines the convex hull [48] of a given set. Considering the nonnegative and sum-to-one constraints, (7) can be regarded as a problem of finding the minimum distance between $\widetilde{\boldsymbol{c}}_i$ and a point $\boldsymbol{x} \in \mathbb{R}^{N_i - 1} \subseteq \mathcal{P}_i$. Therefore, the barycentric coordinate system can be equivalent to the Cartesian coordinate system by converting the convex hull of N_i vertices into the intersection of N_i hyperplanes [48], as shown below

$$\begin{aligned} &\min_{\boldsymbol{x}} \ \|\widetilde{\boldsymbol{c}}_i - \boldsymbol{x}\|_2^2 \\ &\text{s.t. } \boldsymbol{g}_j^{\mathrm{T}} \boldsymbol{x} \leq h_j \quad \forall j \in \mathcal{I}_{N_i} \end{aligned} \tag{8}$$

where $\boldsymbol{g}_j \in \mathbb{R}^{N_i - 1}$ is the normal vector, and h_j is the inner product constant. Through $\boldsymbol{g}_j$ and h_j, it is able to define N_i hyperplanes and convert the problem into N_i linear constraints. Then, we proposed a simplex-based algebraic solution for addressing (8), which can be divided into two cases.

A. Solving (8) When $\widetilde{\boldsymbol{c}}_i \in \mathcal{P}_i$

In the DR space, if the ith SC's spectrum $\widetilde{\boldsymbol{c}}_i$ is already included in the simplex $\mathcal{P}_i$, the minimum distance equals 0 and the optimal $\boldsymbol{x}$ occurs at $\boldsymbol{x}^\star = \widetilde{\boldsymbol{c}}_i$. The optimal weight $\boldsymbol{w}_i^\star \in \mathbb{R}^{N_i}$ can be directly estimated using the following expression:

$$\boldsymbol{w}_i^\star[j] = \frac{h_j - \boldsymbol{g}_j^{\mathrm{T}} \boldsymbol{x}^\star}{h_j - \boldsymbol{g}_j^{\mathrm{T}} \widetilde{\boldsymbol{P}}_i[j]} = \frac{h_j - \boldsymbol{g}_j^{\mathrm{T}} \widetilde{\boldsymbol{c}}_i}{h_j - \boldsymbol{g}_j^{\mathrm{T}} \widetilde{\boldsymbol{P}}_i[j]} \quad \forall j \in \mathcal{I}_{N_i} \tag{9}$$

which is derived from the simple geometrical observations [47]. Specifically, to obtain the jth weight $\boldsymbol{w}_i^\star[j]$, we compare the distances from $\widetilde{\boldsymbol{c}}_i$ and $\widetilde{\boldsymbol{P}}_i[j]$ to the jth hyperplane. In other words, the closer or further $\widetilde{\boldsymbol{c}}_i$ is to $\widetilde{\boldsymbol{P}}_i[j]$, the larger or smaller the weight $\boldsymbol{w}_i^\star[j]$ will be, respectively.

B. Solving (8) When $\widetilde{\boldsymbol{c}}_i \notin \mathcal{P}_i$

When $\widetilde{\boldsymbol{c}}_i$ lies outside the simplex $\mathcal{P}_i$, $\boldsymbol{x}^\star$ is the projected point of $\widetilde{\boldsymbol{c}}_i$ onto the boundary of $\mathcal{P}_i$. As (8) is convex, we adopt the alternating direction method of multipliers (ADMM) [48], a widely used tool in statistical machine learning. Thus, we reformulate (8) as follows:

$$\min_{\boldsymbol{x}} \ \frac{1}{2}\|\widetilde{\boldsymbol{c}}_i - \boldsymbol{x}\|_2^2 + \sum_{j=1}^{N_i} I_{\mathcal{A}_j}(\boldsymbol{x}) \tag{10}$$

where the set $\mathcal{A}_j \triangleq \{\boldsymbol{x} \mid \boldsymbol{g}_j^{\mathrm{T}} \boldsymbol{x} \leq h_j\}$, and $I_{\mathcal{A}_j}(\boldsymbol{x})$ denotes the indicator function of $\mathcal{A}_j$, which is defined as

$$I_{\mathcal{A}_j}(\boldsymbol{x}) \triangleq \begin{cases} 0, & \text{if } \boldsymbol{x} \in \mathcal{A}_j \\ \infty, & \text{otherwise.} \end{cases} \tag{11}$$

For the unconstrained optimization problem (10), we adopt an instance of ADMM—split augmented Lagrangian shrinkage algorithm (SALSA) [49]. One of the essential ingredients in SALSA is the variable splitting (VS) technique, which converts an unconstrained problem into a constrained one by introducing auxiliary variables. By introducing $\boldsymbol{v}_0$ and $\boldsymbol{v}_j$, we can then cast (10) into ADMM form as follows:

$$\begin{aligned} &\min_{\boldsymbol{v}_0, \boldsymbol{v}_j, \boldsymbol{x}} \ \frac{1}{2}\|\widetilde{\boldsymbol{c}}_i - \boldsymbol{v}_0\|_2^2 + \sum_{j=1}^{N_i} I_{\mathcal{A}_j}(\boldsymbol{v}_j) \\ &\text{s.t. } \boldsymbol{v}_0 = \boldsymbol{x}, \ \ \boldsymbol{v}_j = \boldsymbol{x} \quad \forall j \in \mathcal{I}_{N_i}. \end{aligned} \tag{12}$$

The associated augmented Lagrangian of (12) is given by

$$\begin{aligned} &\mathcal{L}(\boldsymbol{v}_0, \boldsymbol{v}_j, \boldsymbol{x}, \boldsymbol{d}_0, \boldsymbol{d}_j) \\ &= \frac{1}{2}\|\widetilde{\boldsymbol{c}}_i - \boldsymbol{v}_0\|_2^2 + \sum_{j \in \mathcal{I}_{N_i}} I_{\mathcal{A}_j}(\boldsymbol{v}_j) \\ &\quad + \frac{\mu}{2}\left(\|\boldsymbol{x} - \boldsymbol{v}_0 - \boldsymbol{d}_0\|_2^2 + \sum_{j \in \mathcal{I}_{N_i}} \|\boldsymbol{x} - \boldsymbol{v}_j - \boldsymbol{d}_j\|_2^2\right) \end{aligned} \tag{13}$$

where $\mu > 0$ is the penalty parameter, and $\boldsymbol{d}_0$ and $\boldsymbol{d}_j$ are scaled dual variables. Then, the ADMM algorithm for solving (12) is summarized in Algorithm 1, and we can obtain $\boldsymbol{x}_i^\star$ in the simplex $\mathcal{P}_i$ and $\boldsymbol{w}_i^\star$ can be directly calculated by (9). Since Algorithm 1 satisfies the convergence condition of linear equality constraints, the above equation guarantees

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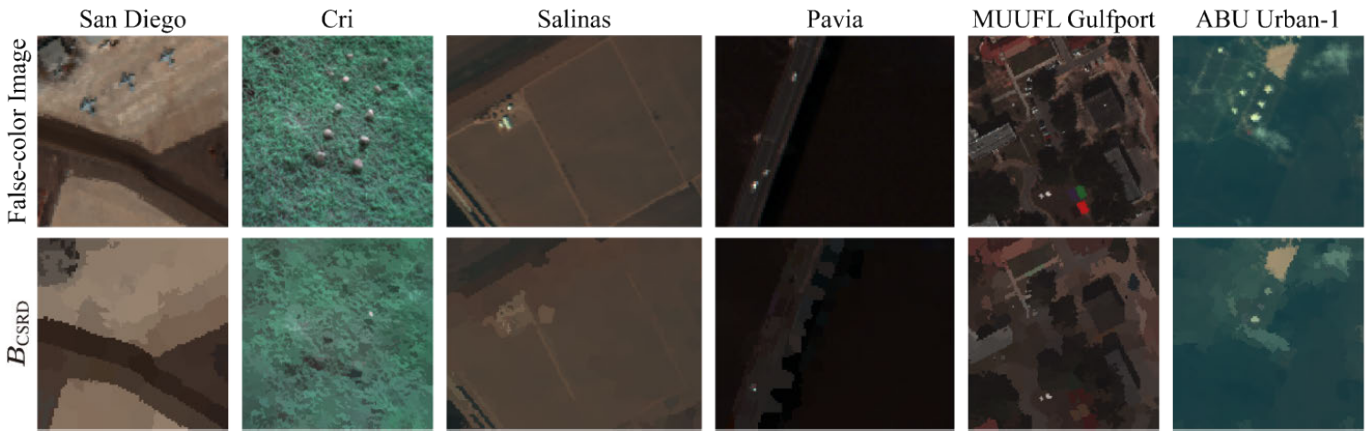


Fig. 4. Comparison of the false-color image and the reconstructed background $\boldsymbol{B}_{\text{CSRD}}$.

Algorithm 1 ADMM Algorithm for Solving (12)

Given $(\widetilde{\boldsymbol{c}}_i, N_i)$, $\mathcal{A}_j$ and $\mu > 0$.
Initialize $\boldsymbol{v}_0^0 = \boldsymbol{v}_j^0 = \boldsymbol{d}_0^0 = \boldsymbol{d}_j^0 := \boldsymbol{0}_{N_i-1}$ (or warm start). Set $t := 0$.
repeat
  Update $(\boldsymbol{v}_0, \{\boldsymbol{v}_j^{t+1}\}_{j\in\mathcal{I}_{N_i}})$ by (28)–(30);
  Update $\boldsymbol{x}^{t+1}$ by (32);
  Update $\boldsymbol{d}_0^{t+1} := \boldsymbol{d}_0^t - \left(\boldsymbol{x}^t - \boldsymbol{v}_0^t\right)$ and $\boldsymbol{d}_j^{t+1} := \boldsymbol{d}_j^t - \left(\boldsymbol{x}^t - \boldsymbol{v}_j^t\right)$;
  $t := t + 1$;
until the predefined stopping criterion is met.
Output $\boldsymbol{x}_i^\star$.

Algorithm 2 ADMM Algorithm for Solving (21)

Given $\boldsymbol{Y}$, $\boldsymbol{B}_{\text{CSRD}}$, and $\rho > 0$.
Obtain $\boldsymbol{E}$ and $\boldsymbol{S}_{\text{CSRD}}$ from $\boldsymbol{B}_{\text{CSRD}}$.
Initialize $\boldsymbol{U}_1^t = \boldsymbol{U}_4^t = \boldsymbol{S}^t = \boldsymbol{J}_1^t = \boldsymbol{J}_4^t := \boldsymbol{0}_{N\times L}$, $\boldsymbol{U}_2^t = \boldsymbol{U}_3^t = \boldsymbol{A}^t = \boldsymbol{J}_2^t = \boldsymbol{J}_3^t := \boldsymbol{0}_{M\times L}$ (or warm start). Set $t := 0$.
repeat
  Update $\left(\boldsymbol{U}_1^{t+1}, \boldsymbol{U}_2^{t+1}, \boldsymbol{U}_3^{t+1}, \boldsymbol{U}_4^{t+1}\right)$ by (34)-(37);
  Update $(\boldsymbol{S}^{t+1}, \boldsymbol{A}^{t+1})$ by (39) and (40);
  Update $\boldsymbol{J}_i^{t+1} = \boldsymbol{J}_i^t - (\boldsymbol{S}^t - \boldsymbol{U}_i^t)$, $\forall i \in \{1, 4\}$ and $\boldsymbol{J}_i^{t+1} = \boldsymbol{J}_i^t - (\boldsymbol{A}^t - \boldsymbol{U}_i^t)$, $\forall i \in \{2, 3\}$;.
  $t := t + 1$;
until the predefined stopping criterion is met.
Output the anomaly matrix $\boldsymbol{A}^\star$.

convergence [48, Lemma 9.2]. For conciseness, we show the whole derivation in Appendix A.

After acquiring $\boldsymbol{w}_i^\star$, the reconstructed spectrum $\boldsymbol{c}_i^\star \in \mathbb{R}^M$ can be defined as

$$\boldsymbol{c}_i^\star = \boldsymbol{P}_i \boldsymbol{w}_i^\star. \tag{14}$$

To revert to the pixel space, we define the reconstructed background $\boldsymbol{B}_{\text{CSRD}} \in \mathbb{R}^{M\times L}$, and $\boldsymbol{B}_{\text{CSRD}}[k]$ is set to $\boldsymbol{c}_i^\star$ if the kth pixel is part of the ith superpixel $\boldsymbol{Y}_i$. In addition, we visualize $\boldsymbol{B}_{\text{CSRD}}$ from each real HSI, as shown in Fig. 4, and the corresponding data descriptions can be found in Section V-A. Interestingly, we observe that the background can approximately fill in the anomalous regions. Since a superpixel segmentation method is introduced into CRD, we term our method CSRD, which is illustrated in Fig. 1(b). Similar to (4), the detection map can be obtained by pixelwisely calculating the difference between the input image $\boldsymbol{Y}$ and the reconstructed background $\boldsymbol{B}_{\text{CSRD}}$.

IV. RPCA With CSRD-Based Regularizer

Although CSRD can roughly conceal the anomalies and quickly reconstruct the background, it may imprecisely characterize anomaly boundaries with a mosaic-like appearance (see Fig. 4), rendering subsequent detection more cumbersome. In recent remote sensing literature [50], it is proven that first obtaining a preliminary rough solution for subsequent regularization not only well solves numerous inverse problems but also avoids math-heavy optimization thereby reducing the computational burden; this elegant thought has been further advanced to the highly challenging meta-hyperspectral imaging in the very recent *Nature* community [51]. Drawing inspiration by Lin et al. [50] and [51], we consider $\boldsymbol{B}_{\text{CSRD}}$ obtained in Section III as a preliminary solution and design it as a regularizer to fully exploit such valuable information.

Different from some previous works [24], [52] that use $\boldsymbol{Y} = \boldsymbol{B} + \boldsymbol{A}$ as a constraint, we further consider the influence of noise in the following form:

$$\boldsymbol{Y} = \boldsymbol{B} + \boldsymbol{A} + \boldsymbol{N} \tag{15}$$

where $\boldsymbol{N}$ denotes the noise matrix. By adding two regularizers in (2), we formulate (15) as a convex optimization problem as

$$\min_{\boldsymbol{B},\boldsymbol{A}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{B} - \boldsymbol{A}\|_F^2 + \lambda_n\|\boldsymbol{B}\|_* + \lambda_a\|\boldsymbol{A}\|_1 + \frac{\lambda_b}{2}\phi(\boldsymbol{B}) \tag{16}$$

where λ_n, λ_a, and λ_b are the regularizer parameters, and ϕ is a customized convex regularizer designed based on CSRD, which is defined as follows:

$$\phi(\boldsymbol{B}) = \|\text{vec}(\boldsymbol{B} - \boldsymbol{B}_{\text{CSRD}})\|_Q^2 = \|\boldsymbol{b} - \boldsymbol{b}_{\text{CSRD}}\|_Q^2 \tag{17}$$

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where $\boldsymbol{b} = \text{vec}(\boldsymbol{B}) \in \mathbb{R}^{ML}$, $\boldsymbol{b}_{\text{CSRD}} = \text{vec}(\boldsymbol{B}_{\text{CSRD}}) \in \mathbb{R}^{ML}$, and $\|\cdot\|_{\boldsymbol{Q}}$ denotes the $\boldsymbol{Q}$-quadratic norm [53], which has been used in hyperspectral inpainting work [50] to extract useful information embedded in unsatisfied small data training results. For $\boldsymbol{Q} \in \mathbb{S}_{++}$ can be any positive semidefinite (PSD) matrix. Given a vector $\boldsymbol{x}$, the so-called $\boldsymbol{Q}$-quadratic norm is defined as $\|\boldsymbol{x}\|_{\boldsymbol{Q}} = \|\boldsymbol{Q}^{(1/2)}\boldsymbol{x}\|_2$. Therefore, we approximately decompose the background as $\boldsymbol{B} \approx \boldsymbol{E}\boldsymbol{S}$ by the eigendecomposition, where $\boldsymbol{E} \in \mathbb{R}^{M\times N}$ denotes the orthonormal basis of the N-dimensional hyperspectral subspace ($N \ll M$), and $\boldsymbol{S} \in \mathbb{R}^{N\times L}$ denotes the corresponding coefficient matrix. By choosing $\boldsymbol{Q} = \boldsymbol{I} \otimes (\boldsymbol{E}\boldsymbol{E}^{\text{T}})$, (17) can be written equivalently as

$$\|\boldsymbol{b} - \boldsymbol{b}_{\text{CSRD}}\|_{\boldsymbol{Q}}^2 = \|\boldsymbol{S} - \boldsymbol{S}_{\text{CSRD}}\|_F^2. \tag{18}$$

In deriving (18), we rely on the fundamental properties of the Kronecker product (see Appendix B). Therefore, (16) can be reformulated as

$$\min_{\boldsymbol{S},\boldsymbol{A}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{E}\boldsymbol{S} - \boldsymbol{A}\|_F^2 + \lambda_n\|\boldsymbol{E}\boldsymbol{S}\|_* + \lambda_a\|\boldsymbol{A}\|_1 + \frac{\lambda_b}{2}\|\boldsymbol{S} - \boldsymbol{S}_{\text{CSRD}}\|_F^2. \tag{19}$$

Since two matrices $\boldsymbol{E}$ and $\boldsymbol{S}$ already possess LR characteristics, as well as $\boldsymbol{E}\boldsymbol{S}$, the nuclear norm in (19) is unnecessary. Instead, a similar purpose (i.e., preserving the LR property of the background) can be achieved just by the CSRD-based regularizer. Hence, (19) can be modified as

$$\min_{\boldsymbol{S},\boldsymbol{A}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{E}\boldsymbol{S} - \boldsymbol{A}\|_F^2 + \lambda_a\|\boldsymbol{A}\|_1 + \frac{\lambda_b}{2}\|\boldsymbol{S} - \boldsymbol{S}_{\text{CSRD}}\|_F^2. \tag{20}$$

To address the unconstrained problem (20), we use SALSA again and achieve separability in the objective function. Consequently, (20) is transformed into a constrained problem by VS as demonstrated below

$$\begin{aligned}\min_{\boldsymbol{U}_1,\boldsymbol{U}_2,\boldsymbol{U}_3,\boldsymbol{U}_4,\boldsymbol{S},\boldsymbol{A}} \ & \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{E}\boldsymbol{U}_1 - \boldsymbol{U}_2\|_F^2 + \lambda_a\|\boldsymbol{U}_3\|_1 \\ & + \frac{\lambda_b}{2}\|\boldsymbol{U}_4 - \boldsymbol{S}_{\text{CSRD}}\|_F^2 \\ \text{s.t. } & \boldsymbol{U}_1 = \boldsymbol{S}, \quad \boldsymbol{U}_2 = \boldsymbol{A}, \ \boldsymbol{U}_3 = \boldsymbol{A}, \ \boldsymbol{U}_4 = \boldsymbol{S}.\end{aligned} \tag{21}$$

The associated augmented Lagrangian of (21) is

$$\begin{aligned}&\mathcal{L}(\boldsymbol{U}_1, \boldsymbol{U}_2, \boldsymbol{U}_3, \boldsymbol{U}_4, \boldsymbol{A}, \boldsymbol{S}, \boldsymbol{J}_1, \boldsymbol{J}_2, \boldsymbol{J}_3, \boldsymbol{J}_4) \\ &= \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{E}\boldsymbol{U}_1 - \boldsymbol{U}_2\|_F^2 + \lambda_a\|\boldsymbol{U}_3\|_1 + \frac{\lambda_b}{2}\|\boldsymbol{U}_4 - \boldsymbol{S}_{\text{CSRD}}\|_F^2 \\ &\quad + \frac{\rho}{2}\big(\|\boldsymbol{S} - \boldsymbol{U}_1 - \boldsymbol{J}_1\|_F^2 + \|\boldsymbol{A} - \boldsymbol{U}_2 - \boldsymbol{J}_2\|_F^2 \\ &\qquad + \|\boldsymbol{A} - \boldsymbol{U}_3 - \boldsymbol{J}_3\|_F^2 + \|\boldsymbol{S} - \boldsymbol{U}_4 - \boldsymbol{J}_4\|_F^2\big)\end{aligned} \tag{22}$$

where $\rho > 0$ is the penalty parameter, and $\boldsymbol{J}_1, \boldsymbol{J}_2, \boldsymbol{J}_3$, and $\boldsymbol{J}_4$ denote the scaled dual variables. The SALSA algorithm is summarized in Algorithm 2. By intuitively observations, (21) meets the convergence condition of linear equality constraints and, as a result, it guarantees convergence [48, Lemma 9.2]. The details can be found in Appendix C. After obtaining the optimal anomaly matrix $\boldsymbol{A}^*$, the final detection result $\boldsymbol{r} \in \mathbb{R}^L$ can be calculated by comprehensively evaluating the spectrum of each pixel in $\boldsymbol{A}^\star$, as shown as

$$\boldsymbol{r}[i] = \|\boldsymbol{A}^\star[i]\|_2. \tag{23}$$

Finally, to enhance the visual effects of the detection result $\boldsymbol{r}$, we design a nonlinear transform called capped-square (CS) transform $f(t)$, where the name is inspired by the famous capped ℓ_1-norm [54]. The function is described as

$$f(t) = \begin{cases} \left(\dfrac{t}{\eta}\right)^2, & t < \eta \\ 1, & t \geq \eta \end{cases} \tag{24}$$

where η denotes a threshold ($\eta := 0.8$ in our work), and the corresponding curve is shown in Fig. 5. Literally, $f(t)$ can be divided into the square part and the capped part by η, which means the background can be gradually depressed while the anomalies can be directly highlighted. Since the superpixel segmentation method is used to accelerate CRD and then combined with the RPCA model, we refer to the proposed algorithm as SuperRPCA, and the pseudocode is summarized in Algorithm 3. Note that all the closed-form solutions do not involve the large inverse matrix calculation, making the calculation time of SuperRPCA faster than other optimization-based methods, which will be demonstrated in Section V-C.

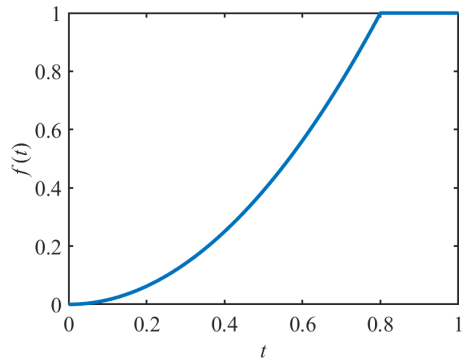


Fig. 5. Function curve of the CS nonlinear transform.

V. Experimental Results

This section aims to showcase the excellence of the proposed SuperRPCA method. The real datasets used for testing are described in Section V-A, while a detailed description of the experimental setup is provided in Section V-B. To demonstrate that the proposed SuperRPCA method has advanced performance, we perform a comprehensive comparison by selecting several benchmark peer methods. Therefore, Section V-C presents qualitative and quantitative analyses. Finally, in Section V-D, we analyze two key parameters in SLIC to verify that the superpixel segmentation results do not need to be precise enough to achieve excellent performance.

A. Data Description

In this section, six real HSIs with distinct characteristics are used to evaluate the effectiveness of the proposed SuperRPCA algorithm, which are described in detail as follows. The first

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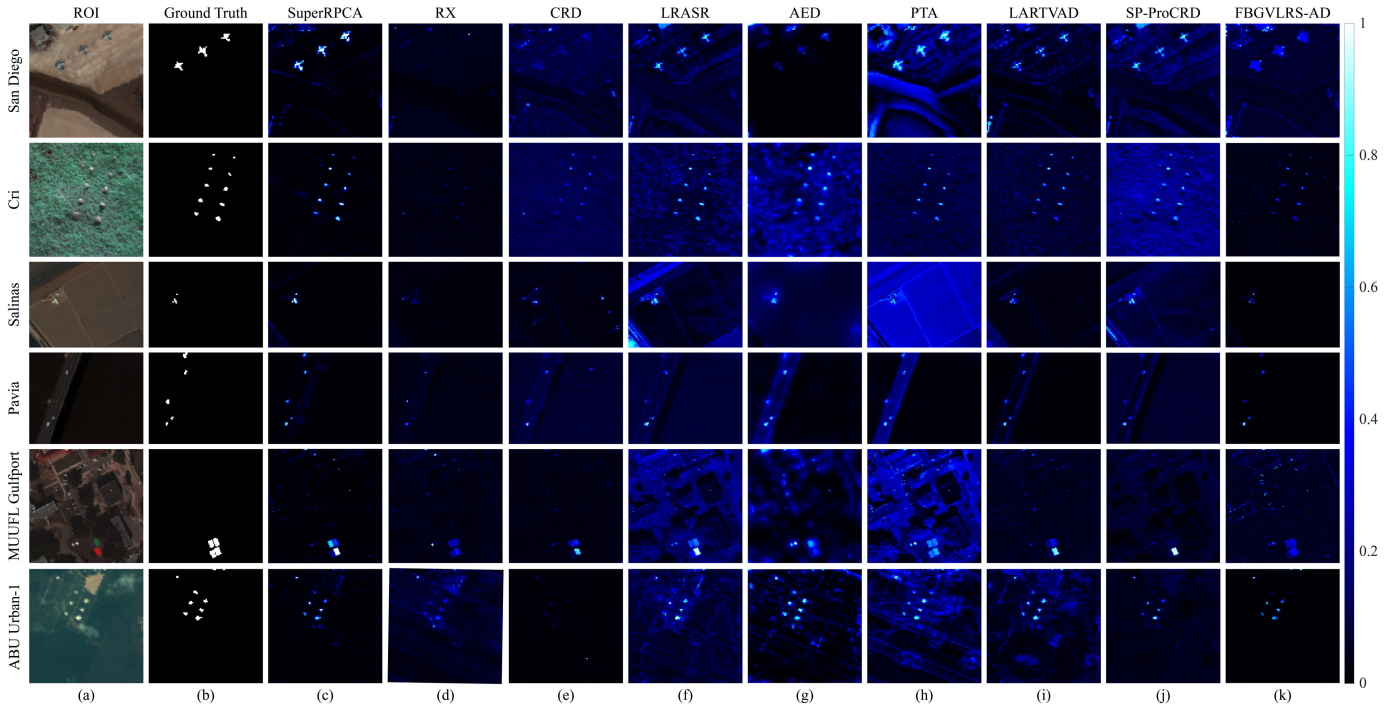


Fig. 6. Qualitative study on each data. (a) False-color image of the region of interest (ROI). (b) Ground truth. Detection results are obtained by (c) proposed SuperRPCA, (d) RX, (e) CRD, (f) LRASR, (g) AED, (h) PTA, (i) LARTVAD, (j) SP-ProCRD, and (k) FBGVLRS-AD.

Algorithm 3 SuperRPCA Algorithm

Given the hyperspectral data $\boldsymbol{Y}$.
Cluster $\boldsymbol{Y}$ as $\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_K$ using SLIC [42].
Obtain $\boldsymbol{c}_i$ from $\boldsymbol{Y}_i$ and then construct $\boldsymbol{C}$ by $\boldsymbol{c}_i$.
Obtain N_i and $\boldsymbol{P}_i$ by search mechanism in Section III.
Obtain $\widetilde{\boldsymbol{C}} = [\widetilde{\boldsymbol{c}}_1, \ldots, \widetilde{\boldsymbol{c}}_K]$ by $\boldsymbol{C}$ and (6).
for $i \leftarrow 1$ to k **do**
  Obtain $\widetilde{\boldsymbol{P}}_i$ from $\widetilde{\boldsymbol{C}}$.
  Compute $\{\mathcal{A}_j, (\boldsymbol{g}_j, h_j)\}_{j=1}^{N_i}$ from $\widetilde{\boldsymbol{P}}_i$.
  if $\widetilde{\boldsymbol{c}}_i \notin \mathcal{P}_i$ **then**
    Obtain $\boldsymbol{x}_i^\star$ by Algorithm 1.
  end if
  Obtain $\boldsymbol{w}_i^\star$ by $\boldsymbol{x}_i^\star$ and (9).
end for
Obtain $\boldsymbol{c}_i^\star$ by and (14).
Obtain $\boldsymbol{B}_{\text{CSRD}}$ by $\boldsymbol{P}_i$ and $\boldsymbol{c}_i^\star$.
Obtain $\boldsymbol{A}^\star$ by Algorithm 2.
Obtain $\boldsymbol{r}$ by (23).
Output detection map $\boldsymbol{r}$.

data, San Diego data [18], are acquired over San Diego, CA, USA, using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor [55], with noisy bands and water absorption bands removed. It comprises $M = 189$ spectral bands and $L = 90 \times 90$ pixels. Anomalies in this dataset correspond to airplanes. The second data, Cri data [18], are captured by the Nuance Cri hyperspectral sensor, with $M = 46$ spectral bands, and $L = 400 \times 400$ pixels, in which the rocks are the anomalies. The third data, Salinas data [56], are collected over Salinas Valley, CA, USA, and are obtained using the AVIRIS sensor. It has $M = 204$ spectral bands and $L = 150 \times 200$ pixels, in which the farmhouses are the anomalies. The fourth data, Pavia data [18], are acquired over Pavia, Italy, using the Reflective Optics System Imaging Spectrometer (ROSIS), with $M = 102$ spectral bands, and $L = 80 \times 100$ pixels. Three vehicles on the bridge and two bare soils near the bridge pier are labeled as anomalies. The fifth data, MUUFL Gulfport data [57], [58], are acquired over Long Beach, MS, USA, using CASI-1500, with $M = 64$ spectral bands, and $L = 150 \times 150$ pixels. Four clothes on the campus of the University of Southern Mississippi are labeled as anomalies. The last data, ABU Urban-1, are acquired from the ABU dataset [59] over Beaumont, TX, USA, using AVIRIS, with $M = 204$ spectral bands, and $L = 100 \times 100$ pixels. Several oil storage tanks in the refinery are labeled as anomalies. For reference, the false-color image and the ground-truth map can be referred to Fig. 6(a) and (b).

B. Experimental Setup

To assess the detection performance of the proposed SuperRPCA, we select eight HAD methods as benchmark methods, including RX [7], CRD [41], LRASR [23], attribute and edge-preserving filtering-based anomaly detector (AED) [59], prior-based tensor approximation (PTA) [60], low average rank with a TV regularization model for HAD (LARTVAD) [52], self-paced probabilistic CRD (SP-ProCRD) [61], and feedback band group and variation LR sparse model (FBGVLRS-AD) [62]. The first three methods have been previously introduced in Section I, while the remaining methods will be introduced subsequently. AED is a morphological-based method and

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consists of two stages, the attribute filter, and the edge-preserving filter. The attribute filter can effectively extract anomaly objects with specific area properties and make full use of the spatial details, and the following edge-preserving filter is regarded as a postprocessing step to smooth the detection map. PTA and LARTVAD are tensor-decomposition-based methods that solve the HAD problem by decomposing data into background and anomaly tensors to retain intrinsic data completeness. For the background components, PTA uses the truncated nuclear-norm and linear TV regularization to impose the global LR and piecewise smoothness properties, respectively, while the $\ell_{2,1}$-norm is applied to anomaly parts to have group sparsity. Similar to PTA, LARTVAD uses TV regularization and $\ell_{2,1}$-norm in the LRR model, and the truncated nuclear norm is replaced by the tensor nuclear-norm regularization. SP-ProCRD is a CRD-based approach that uses the self-paced strategy to simulate the human learning process from simpler to more complex concepts. FBGVLRS-AD adopts an iterative feedback mechanism to dynamically refine the input band group of the LRaSMD model.

Next, we introduce the parameter settings of all the peer methods in detail. The basic tunable parameters of peer methods are set based on the default parameter settings in the reference and we have tried our best to optimize the performance; besides, the randomness of methods is fixed. For the inner window w_{in} and outer window w_{out} used in CRD, the search ranges are 3–29 and 5–29, respectively, with intervals of 2. In addition, two regularization parameters involved in LRASR, β_{LRASR} and λ_{LRASR}, are chosen from $\{0.001, 0.01, 0.1, 0.5, 1\}$ and $\{0.001, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$, respectively. Refer to LRASR, we set the number of clusters for the background dictionary to 15, and 20 pixels are selected in each cluster as atoms. Regarding AED, the DR number M and the threshold κ are decided from $\{1, 3, 5, 10, 20\}$ and $\{5, 15, 25, 50, 100, 200, 400\}$, respectively. In the edge-preserving filtering step, two filter parameters σ_s and σ_r are chosen from $\{1, 5, 10, 20, 30, 40, 50\}$ and $\{0.1, 0.2, 0.5, 1, 2, 5\}$, respectively. To select the truncate rank r_{PTA} for PTA, a range from 1 to 5 is considered. In addition, two regularization parameters, α_{PTA} and β_{PTA}, are chosen from $\{0.001, 0.01, 0.1, 1, 10\}$. Two penalty parameters, τ_{PTA} and μ_{PTA}, are also selected from $\{0.001, 0.01, 0.1, 1, 10\}$ and $\{0.0001, 0.001, 0.01\}$, respectively. For LARTVAD, both the regularization parameters β_{LARTVAD} and λ_{LARTVAD}, are explored from $\{0.01, 0.1, 1, 2, 3, 10\}$. Due to the use of the LRR model, the cluster number K_{LARTVAD} is searched from 60 to 200 with an interval of 20 for dictionary construction. For SP-ProCRD, the number of selected bands is set to ten, the second bandwidth is selected from 0.05 to 0.3, the remaining parameters are fixed as $\alpha_{\text{SP-ProCRD}} = \beta_{\text{SP-ProCRD}} := 0.01$, and $\lambda_{\text{SP-ProCRD}} := 0.1$. For FBGVLRS-AD, p is considered within the range 9–15, and m is selected from 2 to 9. Finally, for the proposed SuperRPCA, two regularization parameters, λ_a and λ_b, can be set as 1 and 0.1 ($\lambda_b := 1$ in San Diego data), which will be discussed in Section V-D. In addition, N determines the extracted subspace number of the CSRD rough reconstructed background, and we suggest that N is set no larger than 5.

C. Qualitative and Quantitative Assessment

In this section, the proposed SuperRPCA is evaluated by comparing it with the aforementioned benchmark methods. For the detection results of all the methods, a lighter/darker color means that the pixels are detected as anomalous/background. For RX, the visual effect of each data is overall dark, as can be seen from Fig. 6(d). For CRD, the anomalies can be slightly observed in Cri, but they are wrongly detected or even missed on the remaining data, as seen in Fig. 6(e). It is worth noting that CRD uses a double concentric sliding window strategy, so they are less able to handle cases when anomalies exist with different sizes in the same scene. Given the requirements of constructing a dictionary to represent the background in LRASR, we observe that the dictionary cannot be effectively constructed for the Salinas data, MUUFL Gulfport data, and ABU Urban-1 data [see Fig. 6(f)], and some parts of the images are considered anomalous. Besides, compared with CRD, the visual effects are more prominent in LRASR and PTA [see Fig. 6(f) and (h)]; however, the backgrounds cannot be depressed well by PTA, especially in the San Diego data, Salinas data, and MUUFL Gulfport data. For AED, most of the data have an oversmoothing phenomenon because of the edge-preserving filter, as shown in Fig. 6(g). Two obvious instances happened in the Salinas data and MUUFL Gulfport data. In the Salinas data, three anomaly objects are present, but only two are detected. Similarly, in the MUUFL Gulfport data, while four clothes are identified, their boundaries are unclear. These examples highlight AED's struggle in accurately representing closely spaced anomalies. Although LARTVAD has a better performance in background depressing and anomaly highlighting, the appearance of the anomalies is imprecise in San Diego data in Fig. 6(i). For SP-ProCRD, several roads are detected in Salinas data, but the anomalies can be precisely positioned in Fig. 6(j). For FBGVLRS-AD, the detection results of the San Diego data and MUUFL Gulfport data are multiplied by ten for better visual effects; however, it appears that the method struggles with detecting large anomalies [see Fig. 6(k)]. Instead, we find that the proposed SuperRPCA greatly reduces the wrong detection on the background meanwhile keeping the appearance similar to the ground-truth detection map [see Fig. 6(c)]. For example, in the San Diego and Salinas data, the appearance of the anomalies (i.e., planes and farmhouses) can be correctly characterized, which outperforms other methods.

To fairly assess the proposed SuperRPCA and the peer methods, we use the receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) metrics [63]. The ROC curve is widely used to analyze the performance of the detectors which is composed of the probability of detection P_d and false alarm rate P_f, as shown as

$$P_d = \frac{\text{TP}}{P}, \quad P_f = \frac{\text{FP}}{N} \tag{25}$$

where TP, FP, P, and N represent the counts of pixels classified as anomalies correctly, the counts of pixels that are actually background but incorrectly identified as anomalies, the total number of anomaly pixels, and the total number of background pixels, respectively. A good detector will have a high P_d meanwhile possessing a low P_f; therefore, the

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TABLE I
QUANTITATIVE PERFORMANCE OF THE PROPOSED SUPERRPCA AND BASELINES USING TESTING DATA

Data	Metrics	RX	CRD	LRASR	AED	PTA	LARTVAD	SP-ProCRD	FBGVLRS-AD	SuperRPCA
San Diego	AUC	0.9106	0.9684	0.9541	0.9814	0.9946	0.9900	0.9861	0.9877	**0.9969**
	Time (sec.)	**0.0435**	101.3170	19.8880	0.2295	16.3107	33.9760	1.5596	9.1004	8.4745
Cri	AUC	0.9675	0.9526	0.9848	0.9910	0.9925	0.9642	0.9741	0.9638	**0.9992**
	Time (sec.)	**0.1332**	265.1084	362.0233	1.5622	74.0143	896.0799	54.3418	63.8996	38.3722
Salinas	AUC	0.9994	0.9958	0.9946	0.9998	0.9982	0.9998	0.9994	0.9979	**0.9999**
	Time (sec.)	**0.1316**	321.8022	92.1175	0.9631	62.6503	142.5437	11.2659	55.0898	35.4056
Pavia	AUC	0.9984	0.9969	0.9767	0.9977	0.9995	0.9994	0.9982	0.9994	**0.9997**
	Time (sec.)	**0.0242**	159.5136	14.9021	0.3351	7.0255	21.1801	1.2464	5.9762	4.2539
MUUFL Gulfport	AUC	0.9948	0.9796	0.9957	0.9807	0.9887	0.9950	0.9825	0.9900	**0.9963**
	Time (sec.)	**0.0337**	27.5119	53.9512	0.5351	16.261	63.0724	16.3701	6.7889	8.7760
ABU Urban-1	AUC	0.9907	0.9948	0.9751	**0.9979**	0.9917	0.9922	0.9868	0.9966	0.9976
	Time (sec.)	**0.0491**	46.1141	23.6367	0.7644	24.5054	42.8924	6.9309	15.9761	12.7861
Average	AUC	0.9769	0.9814	0.9802	0.9914	0.9942	0.9901	0.9878	0.9892	**0.9983**
	Time (sec.)	**0.0692**	153.5612	94.4198	0.7316	33.4612	199.9574	15.2858	26.1385	18.0114

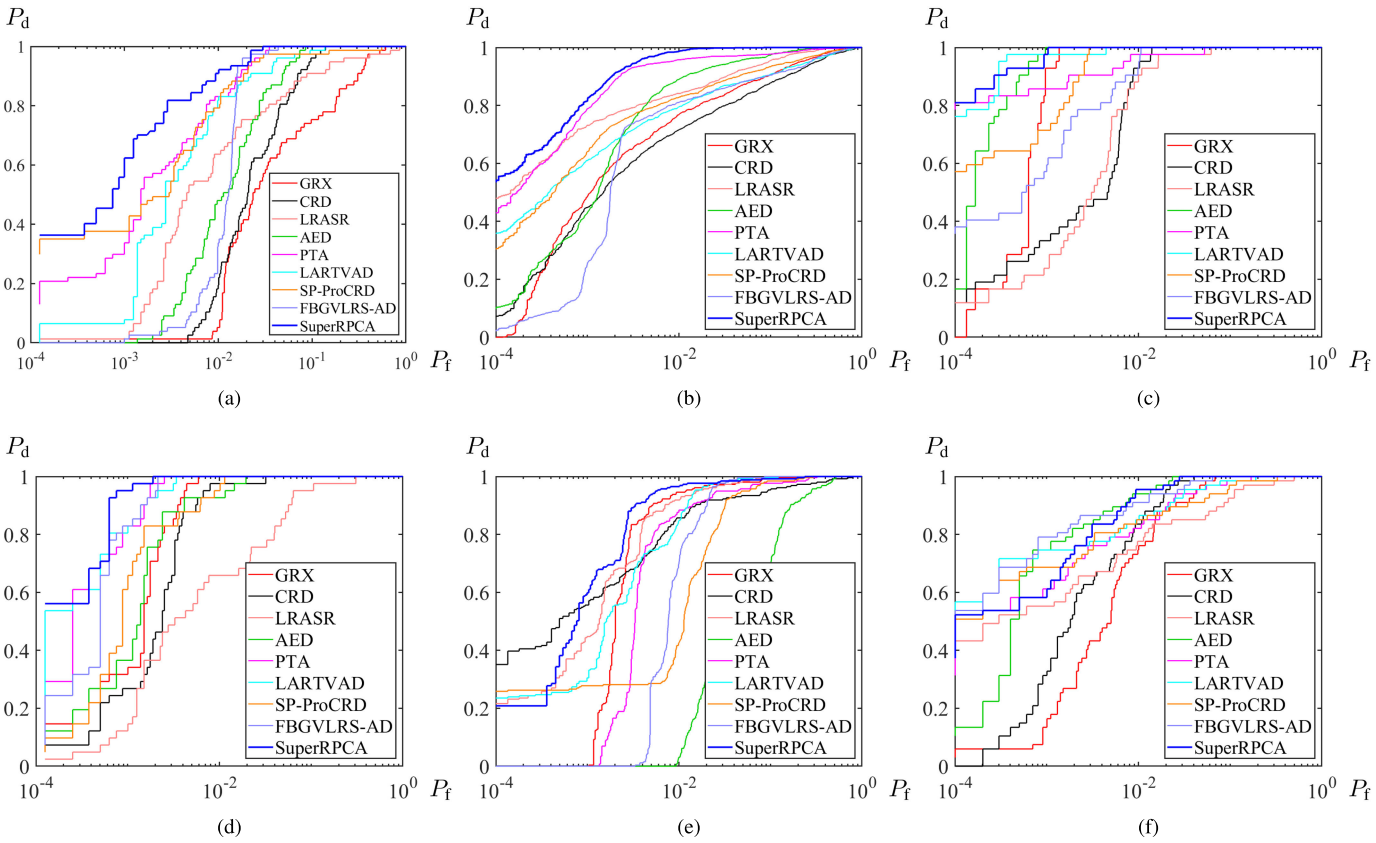


Fig. 7. ROC curves in logarithmic scale of the proposed SuperRPCA and the compared methods. (a) San Diego data. (b) Cri data. (c) Salinas data. (d) Pavia data. (e) MUUFL Gulfport data. (f) ABU Urban-1 data.

ROC curve will be closer to the northwest. Furthermore, the separability maps are also plotted as visual aids. Intuitively, the farther the two boxes are apart, the clearer to distinguish the anomaly and the background. Finally, the computational time is used to evaluate the computational complexity; the experiments for all the methods were conducted and evaluated on a computer that has an Intel Core-i7-9700k CPU with a speed of 3.60 GHz and 32-GB RAM. The implementation of all the methods was carried out in Mathworks MATLAB based on their executable version.

The ROC curve and the separability map are shown in Figs. 7 and 8, respectively. The overall quantitative indices are summarized in Table I. The boldfaced indices are the best values for each quantitative index (i.e., the highest AUC and the fastest computational time), and the second values are underscored. As it turns out, SP-ProCRD performs nearly the best on the Salinas data (AUC = 0.9994), while FBGVLRS-AD gets the third place (but still nearly the best) on the ABU Urban-1 data (AUC = 0.9966), as can be seen from Table I. SuperRPCA outperforms all other methods in terms of AUC for five out of six data. While AED achieves a marginally higher AUC on the ABU Urban-1 dataset, SuperRPCA remains highly competitive due to good background suppression [see Fig. 6(c)], which might be beneficial for

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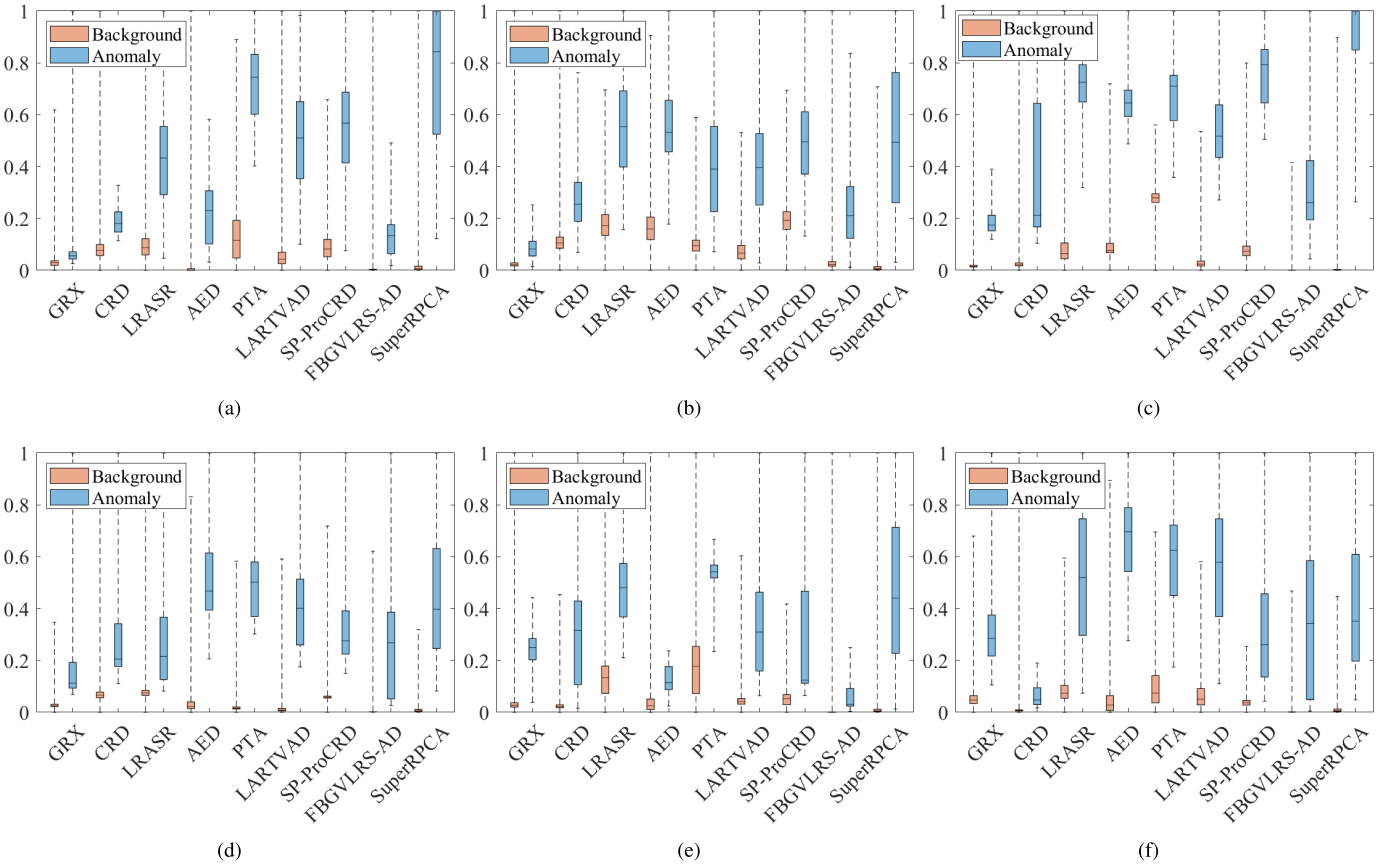


Fig. 8. Separability maps of the proposed SuperRPCA and the compared methods. (a) San Diego data. (b) Cri data. (c) Salinas data. (d) Pavia data. (e) MUUFL Gulfport data. (f) ABU Urban-1 data.

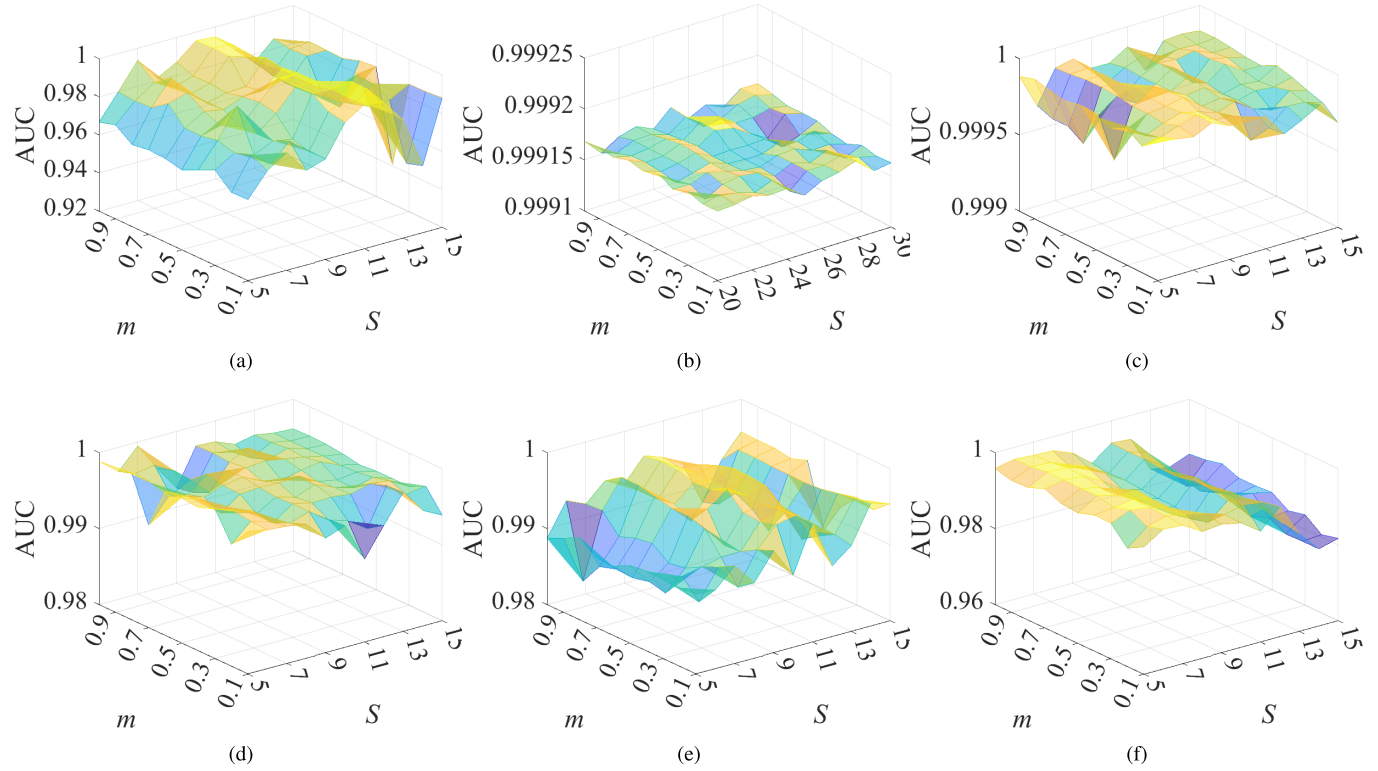


Fig. 9. Sensitivity to SLIC parameters of the proposed SuperRPCA. (a) San Diego data. (b) Cri data. (c) Salinas data. (d) Pavia data. (e) MUUFL Gulfport data. (f) ABU Urban-1 data.

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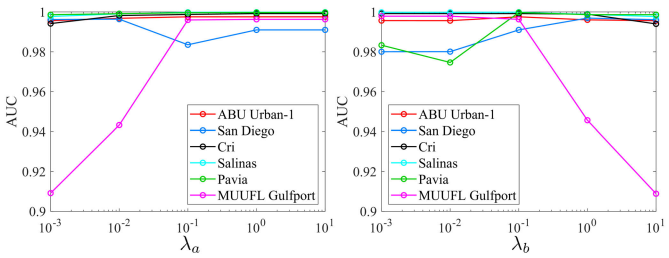


Fig. 10. Impact of λ_a (λ_b is fixed to 0.1) and λ_b (λ_a is fixed to 1) over AUC on each data.

accurate anomaly identification. Regarding the separability map, SuperRPCA pulls the two boxes apart well, especially in Fig. 8(a) and (c). In terms of time index, given the fact that RX merely calculates the Mahalanobis distance between each pixel and the whole image, it has the fastest computational time. CRD and SP-ProCRD also use a pixelwise alculation mechanism, but the large number of dictionary atoms of some data results in high time costs. For LRASR, it takes lots of time to solve the nuclear norm in each optimization iteration. A similar situation also happens with PTA and LARTVAD, which face truncated nuclear norm and tensor nuclear norm, respectively. Despite being the fourth fastest of SuperRPCA, our method gets the first rank among the optimization-based peer methods (including LRASR, PTA, LARTVAD, and FBGVLRS-AD). It means that our algorithm attains precise detection while maintaining high computational efficiency.

D. Model-Order Selection

First, we show the stability without strictly setting SLIC parameters. In other words, even when the reconstructed background $\boldsymbol{B}_{\text{CSRD}}$ is less precise, valuable information can still be extracted under $\boldsymbol{Q}$-quadratic regularization. Fig. 9 illustrates the AUC when jointly considering the two SLIC parameters. Here, S is chosen from 5 to 15 with an increment of 1 (Cri data is chosen from 20 to 30 due to its data scale), while m is varied from 0.1 to 1 with an increment of 0.1. As seen in Fig. 9(a) and (f), the proposed SuperRPCA only exhibits a very slight sensitivity to S for the San Diego data and ABU Urban-1 data. On the contrary, the AUC for the other four data shows an extremely low sensitivity to the variations in S and m. Besides, in contrast to LRX and CRD, which require precise adjustment of the inner and outer window sizes based on anomaly size, SuperRPCA exhibits resilience to deviations in superpixel size selection. This characteristic renders SuperRPCA a more practical and user-friendly tool across diverse tasks.

For the proposed SuperRPCA, two regularization parameters, λ_a and λ_b, need to be properly set. Therefore, we test SuperRPCA across a wide range of λ_a and λ_b to evaluate the quality of the detection performance in terms of AUC. Specifically, both λ_a and λ_b are chosen from $\{0.001, 0.01, 0.1, 1, 10\}$, and the experiments are conducted by varying one parameter individually while keeping the other parameter fixed. The results are summarized in Fig. 10, and we suggest future users of SuperRPCA use the default settings of $\lambda_a = 1$ and $\lambda_b = 0.1$,

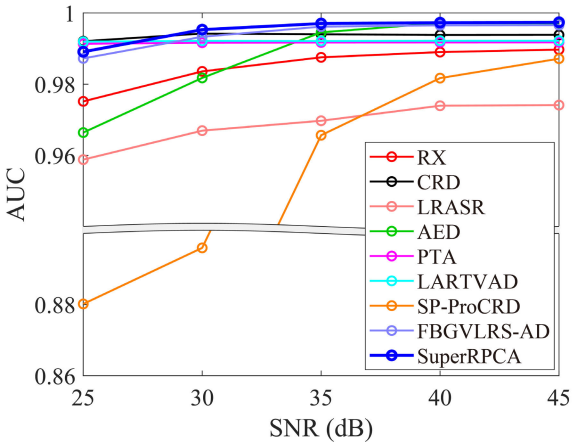


Fig. 11. AUC values of different methods on the ABU Urban-1 data with SNRs 25, 30, 35, 40, and 45 dB.

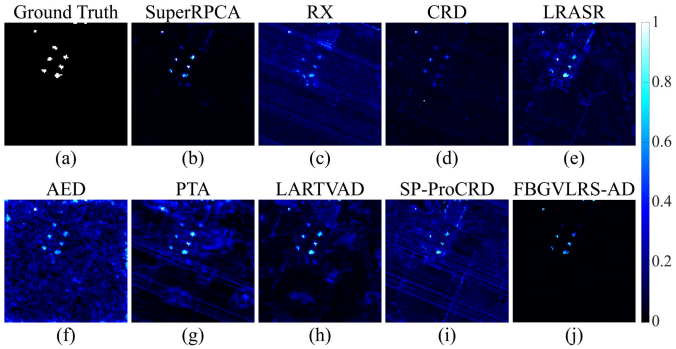


Fig. 12. Qualitative performance on ABU Urban-1 data with SNR 30 dB. (a) Ground truth. Detection results are obtained by (b) proposed SuperRPCA, (c) RX, (d) CRD, (e) LRASR, (f) AED, (g) PTA, (h) LARTVAD, (i) SP-ProCRD, and (j) FBGVLRS-AD.

which have been used throughout most of the experiments in this article.

E. Interference Testing

During image acquisition, remote sensing data are often corrupted by diverse sources of degradation, variability, and noise. Section V-C uses data from diverse hyperspectral sensors with varying spatial resolutions, acquired across different locations and times. The inherent variety in these data validates SuperRPCA's robustness to spatial degradation and spectral variability, as evidenced by its state-of-the-art detection performance. Therefore, in this section, we focus on the effect of noise interference.

We evaluate SuperRPCA's performance against peer methods on the ABU Urban-1 data. The data were corrupted with zero-mean additive Gaussian noise at various SNRs of 25, 30, 35, 40, and 45 dB. The corresponding AUC values are organized in Fig. 11. While its AUC began to decrease slightly at 25-dB SNR, it consistently achieved the highest AUC at the remaining noise levels. Besides, we present the qualitative results on the data with 30-dB SNR, as can be seen from Fig. 12. We point out that while AED slightly outperforms SuperRPCA on the noise-free ABU Urban-1 data (see Fig. 6(g) and Table I), SuperRPCA significantly

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outperforms it when noise is added, meaning that SuperRPCA has sufficient capability to resist the noise interference.

VI. Conclusion

We proposed a novel algorithm, named SuperRPCA, for anomaly detection in HSIs. The algorithm comprises two primary stages. In the first stage, a roughly reconstructed background is obtained using a superpixel-enhanced variant of the CRD, which is termed CSRD. In addition, a simplex-based algebraic solution is proposed for solving CSRD from a geometric perspective. In the second stage, the reconstructed background serves as a pivotal prior and can be formulated as a convex regularizer, subsequently incorporated into the RPCA optimization stage. Using a customized nonlinear CS transform, the background can be suppressed while enhancing the distinguishability of anomalies in the detection result. Remarkably, the designed regularizer is capable of replacing the time-consuming nuclear norm, resulting in a simpler and more efficient algorithm; in the future, quantum deep-learning-based [64] real-time HAD is our ongoing research. We conduct extensive experiments on six real hyperspectral datasets and compare the performance of our proposed SuperRPCA with the existing benchmark peer methods. The results demonstrate that SuperRPCA achieves state-of-the-art accuracy in anomaly detection and provides exceptional qualitative performance through detection maps that clearly distinguish anomaly objects and accurately delineate their boundaries.

Appendix

A. Derivation of Algorithm 1

1) Solution for Line 4 in Algorithm 1: For variables $\boldsymbol{v}_0$ and $\boldsymbol{v}_j$, they can be decoupled as

$$\boldsymbol{v}_0^{t+1} \in \arg\min_{\boldsymbol{v}_0} \frac{1}{2}\|\widetilde{\boldsymbol{c}}_i - \boldsymbol{v}_0\|_2^2 + \frac{\mu}{2}\left\|\boldsymbol{x}^t - \boldsymbol{v}_0 - \boldsymbol{d}_0^t\right\|_2^2 \tag{26}$$

$$\boldsymbol{v}_j^{t+1} \in \arg\min_{\boldsymbol{v}_j \in \mathcal{A}_j} \frac{\mu}{2}\left\|\boldsymbol{v}_j - \boldsymbol{n}_j^t\right\|_2^2 \tag{27}$$

where $\boldsymbol{n}_j = \boldsymbol{x} - \boldsymbol{d}_j$, and the superscript "$t$" denotes the iteration number rather than their power. The closed-form solution of (26) is, respectively, given by

$$\boldsymbol{v}_0^{t+1} = \frac{\widetilde{\boldsymbol{c}}_i + \mu\left(\boldsymbol{x}^t - \boldsymbol{d}_0^t\right)}{\mu + 1}. \tag{28}$$

For the closed-form solution of (27), it can be divided into two cases

1) *Case 1:* If $\boldsymbol{n}_j^t \in \mathcal{A}_j$, the minimum distance equals 0 and exists in

$$\boldsymbol{v}_j^{t+1} = \boldsymbol{n}_j^t. \tag{29}$$

2) *Case 2:* If $\boldsymbol{n}_j^t \notin \mathcal{A}_j$, it means that we need to project $\boldsymbol{n}_j^t$ onto the boundary [48] of the set $\mathcal{A}_j$

$$\boldsymbol{v}_j^{t+1} = \boldsymbol{n}_j^t - \frac{\left(\boldsymbol{g}_j^{\mathrm{T}}\boldsymbol{n}_j^t - h_j\right)\boldsymbol{g}_j}{\|\boldsymbol{g}_j\|^2}. \tag{30}$$

2) Solution for Line 5 in Algorithm 1: The problem can be formulated as

$$\boldsymbol{x}^{t+1} \in \arg\min_{\boldsymbol{x}} \left\|\boldsymbol{x} - \boldsymbol{v}_0^t - \boldsymbol{d}_0^t\right\|_2^2 + \sum_{j\in\mathcal{I}_{N_i}} \left\|\boldsymbol{x} - \boldsymbol{v}_j^t - \boldsymbol{d}_j^t\right\|_2^2 \tag{31}$$

and the following closed-form solution is derived as

$$\boldsymbol{x}^{t+1} = \frac{\boldsymbol{v}_0^t + \boldsymbol{d}_0^t + \sum_{j\in\mathcal{I}_{N_i}}\left(\boldsymbol{v}_j^t + \boldsymbol{d}_j^t\right)}{N_i + 1}. \tag{32}$$

B. Derivation of (18)

The derivation of (18) can be detailed as follows:

$$\begin{aligned}
&\|\boldsymbol{b} - \boldsymbol{b}_{\mathrm{CSRD}}\|_{\boldsymbol{Q}}^2 \\
&\quad= (\boldsymbol{b} - \boldsymbol{b}_{\mathrm{CSRD}})^{\mathrm{T}}\left[\boldsymbol{I}_L \otimes \left(\boldsymbol{E}\boldsymbol{E}^{\mathrm{T}}\right)\right](\boldsymbol{b} - \boldsymbol{b}_{\mathrm{CSRD}}) \\
&\quad= (\boldsymbol{b} - \boldsymbol{b}_{\mathrm{CSRD}})^{\mathrm{T}}(\boldsymbol{I}_L \otimes \boldsymbol{E})\left(\boldsymbol{I}_L \otimes \boldsymbol{E}^{\mathrm{T}}\right)(\boldsymbol{b} - \boldsymbol{b}_{\mathrm{CSRD}}) \\
&\quad= \left\|\left(\boldsymbol{I}_L \otimes \boldsymbol{E}^{\mathrm{T}}\right)(\boldsymbol{b} - \boldsymbol{b}_{\mathrm{CSRD}})\right\|_2^2 \\
&\quad= \left\|\mathrm{vec}\left[\boldsymbol{E}^{\mathrm{T}}(\boldsymbol{B} - \boldsymbol{B}_{\mathrm{CSRD}})\boldsymbol{I}_L\right]\right\|_2^2 \\
&\quad= \|\boldsymbol{E}^{\mathrm{T}}\boldsymbol{E}(\boldsymbol{S} - \boldsymbol{S}_{\mathrm{CSRD}})\|_F^2 \\
&\quad= \|\boldsymbol{S} - \boldsymbol{S}_{\mathrm{CSRD}}\|_F^2.
\end{aligned}$$

The second and fourth equations are the key steps, corresponding to the two basic properties of the Kronecker product, namely, $\boldsymbol{A}\boldsymbol{C}\otimes\boldsymbol{B}\boldsymbol{D} = (\boldsymbol{A}\otimes\boldsymbol{B})(\boldsymbol{C}\otimes\boldsymbol{D})$ and $(\boldsymbol{C}^{\mathrm{T}}\otimes\boldsymbol{A})\mathrm{vec}(\boldsymbol{B}) = \mathrm{vec}(\boldsymbol{A}\boldsymbol{B}\boldsymbol{C})$.

C. Derivation of Algorithm 2

1) Solution for Line 5 in Algorithm 2: Note that $\boldsymbol{U}_1$ and $\boldsymbol{U}_2$ are coupled in the first term of (22), so we need to solve them jointly as shown as

$$\begin{aligned}
&\boldsymbol{U}_1^{t+1}, \boldsymbol{U}_2^{t+1} \\
&\quad\in \arg\min_{\boldsymbol{U}_1,\boldsymbol{U}_2} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{E}\boldsymbol{U}_1 - \boldsymbol{U}_2\|_F^2 \\
&\qquad+ \frac{\rho}{2}\left(\left\|\boldsymbol{S}^t - \boldsymbol{U}_1 - \boldsymbol{J}_1^t\right\|_F^2 + \left\|\boldsymbol{A}^t - \boldsymbol{U}_2 - \boldsymbol{J}_2^t\right\|_F^2\right).
\end{aligned} \tag{33}$$

We define a new variable $\boldsymbol{U}_{12} = [\boldsymbol{U}_1^{\mathrm{T}}, \boldsymbol{U}_2^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{(M+N)\times L}$, and (33) can be reformulated as

$$\boldsymbol{U}_{12}^{t+1} \in \arg\min_{\boldsymbol{U}_{12}} \frac{1}{2}\|\boldsymbol{Y} - \boldsymbol{K}\boldsymbol{U}_{12}\|_F^2 + \frac{\rho}{2}\|\boldsymbol{R}^t - \boldsymbol{U}_{12}\|_F^2 \tag{34}$$

where $\boldsymbol{K} = [\boldsymbol{E}, \boldsymbol{I}_M] \in \mathbb{R}^{M\times(M+N)}$, and $\boldsymbol{R} = [(\boldsymbol{S} - \boldsymbol{J}_1)^{\mathrm{T}}, (\boldsymbol{A} - \boldsymbol{J}_2)^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{(M+N)\times L}$. A closed-form solution of (33) is

$$\boldsymbol{U}_{12}^{t+1} = \left(\boldsymbol{K}^{\mathrm{T}}\boldsymbol{K} + \rho\boldsymbol{I}_{M+N}\right)^{-1}\left(\boldsymbol{K}^{\mathrm{T}}\boldsymbol{Y} + \rho\boldsymbol{R}^t\right). \tag{35}$$

On the contrary, the closed-form solution of $\boldsymbol{U}_3$ and $\boldsymbol{U}_4$ can be, respectively, derived

$$\begin{aligned}
\boldsymbol{U}_3^{t+1} &\in \arg\min_{\boldsymbol{U}_3} \lambda_a\|\boldsymbol{U}_3\|_1 + \frac{\rho}{2}\left\|\boldsymbol{A}^t - \boldsymbol{U}_3 - \boldsymbol{J}_3^t\right\|_F^2 \\
&= \mathrm{shrinkage}\left(\boldsymbol{A}^t - \boldsymbol{J}_3^t, \frac{\lambda_a}{\rho}\right)
\end{aligned} \tag{36}$$

$$\begin{aligned}
\boldsymbol{U}_4^{t+1} &\in \arg\min_{\boldsymbol{U}_4} \frac{\lambda_b}{2}\|\boldsymbol{U}_4 - \boldsymbol{S}_{\mathrm{CSRD}}\|_F^2 + \frac{\rho}{2}\left\|\boldsymbol{S}^t - \boldsymbol{U}_4 - \boldsymbol{J}_4^t\right\|_F^2 \\
&= \frac{\lambda_b\boldsymbol{S}_{\mathrm{CSRD}} + \rho\left(\boldsymbol{S}^t - \boldsymbol{J}_4^t\right)}{\lambda_b + \rho}
\end{aligned} \tag{37}$$

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where shrinkage(x, λ) is the shrinkage operator [65]. It is defined as

$$\text{shrinkage}\big(x, \lambda\big) = \text{sgn}(x)\max\big(|x| - \lambda, 0\big) \tag{38}$$

where sgn$(\cdot)$ is the sign function. Note that we define the matrix version of (38) and apply the operation to each element individually.

2) Solution for Line 6 in Algorithm 2: Due to the separability of two variables $\boldsymbol{S}$ and $\boldsymbol{A}$ in line 6, the simple closed-form solutions are, respectively, shown as

$$\begin{aligned}\boldsymbol{S}^{t+1} &\in \arg\min_{\boldsymbol{S}} \ \frac{\rho}{2}\left(\left\|\boldsymbol{S} - \boldsymbol{U}_1^t - \boldsymbol{J}_1^t\right\|_F^2 + \left\|\boldsymbol{S} - \boldsymbol{U}_4^t - \boldsymbol{J}_4^t\right\|_F^2\right) \\ &= \frac{\boldsymbol{U}_1^t + \boldsymbol{J}_1^t + \boldsymbol{U}_4^t + \boldsymbol{J}_4^t}{2} \end{aligned} \tag{39}$$

$$\begin{aligned}\boldsymbol{A}^{t+1} &\in \arg\min_{\boldsymbol{A}} \ \frac{\rho}{2}\left(\left\|\boldsymbol{A} - \boldsymbol{U}_2^t - \boldsymbol{J}_2^t\right\|_F^2 + \left\|\boldsymbol{A} - \boldsymbol{U}_3^t - \boldsymbol{J}_3^t\right\|_F^2\right) \\ &= \frac{\boldsymbol{U}_2^t + \boldsymbol{J}_2^t + \boldsymbol{U}_3^t + \boldsymbol{J}_3^t}{2}. \end{aligned} \tag{40}$$

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Jhao-Ting Lin (Student Member, IEEE) received the B.S. degree from the Department of Communications, Navigation and Control Engineering, National Taiwan Ocean University, Keelung, Taiwan, in 2020. He is currently pursuing the Ph.D. degree with the Intelligent Hyperspectral Computing Laboratory, Department of Electrical Engineering, National Cheng Kung University, Tainan, Taiwan.

His research interests include convex optimization, signal processing, anomaly detection, and hyperspectral imaging.

Mr. Lin has just received a highly competitive scholarship, i.e., the Pan Wen Yuan Award, from the Industrial Technology Research Institute (ITRI) of Taiwan, in 2022.



Chia-Hsiang Lin (Member, IEEE) received the B.S. degree in electrical engineering and the Ph.D. degree in communications engineering from National Tsing Hua University (NTHU), Hsinchu, Taiwan, in 2010 and 2016, respectively.

From 2015 to 2016, he was a Visiting Student at Virginia Tech, Arlington, VA, USA. He is currently an Associate Professor with the Department of Electrical Engineering and Miin Wu School of Computing, National Cheng Kung University (NCKU), Tainan, Taiwan. Before joining NCKU, he held research positions with The Chinese University of Hong Kong, Hong Kong, in 2014 and 2017, NTHU from 2016 to 2017, and the University of Lisbon (ULisboa), Lisbon, Portugal, from 2017 to 2018. He was an Assistant Professor at the Center for Space and Remote Sensing Research, National Central University, Taoyuan, Taiwan, in 2018, and a Visiting Professor at ULisboa, in 2019. His research interests include network science, quantum computing, convex geometry and optimization, signal processing, and imaging science.

Dr. Lin received the Emerging Young Scholar Award from the National Science and Technology Council (NSTC), in 2023, the Future Technology Award from NSTC, in 2022, the Outstanding Youth Electrical Engineer Award from The Chinese Institute of Electrical Engineering (CIEE), in 2022, the Best Young Professional Member Award from IEEE Tainan Section, in 2021, the Prize Paper Award from IEEE Geoscience and Remote Sensing Society (GRS-S), in 2020, the Top Performance Award from Social Media Prediction Challenge at ACM Multimedia, in 2020, and The Third Place from AIM Real World Super-Resolution Challenge at IEEE International Conference on Computer Vision (ICCV), in 2019. He received the Ministry of Science and Technology (MOST) Young Scholar Fellowship, together with the EINSTEIN Grant Award, from 2018 to 2023. In 2016, he was a recipient of the Outstanding Doctoral Dissertation Award from the Chinese Image Processing and Pattern Recognition Society and the Best Doctoral Dissertation Award from the IEEE GRS-S.

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